

ULTRASOUND INSTRUMENTATION

OBJECTIVES 332

PRESENTATION MODES 332

A-Mode 332

B-Mode 332

M-Mode 333

TIME REQUIRED TO OBTAIN IMAGES 333

SYSTEM COMPONENTS 335

SIGNAL PROCESSING 335

Preprocessing 335

Postprocessing 336

DYNAMIC RANGE 337

ULTRASOUND IMAGE ARTIFACTS 338

Speckle 338

Shadowing and Enhancement 338

Multiple Pathway 338

Refraction 338

QUALITY CONTROL 338

Quality Control Tests 339

PROBLEMS 340

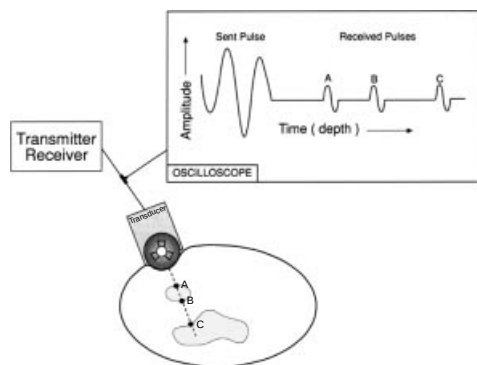
SUMMARY 341

REFERENCES 341

OBJECTIVES

After completing this chapter, the reader should be able to:

- Name three modes of ultrasound imaging.
- Determine the time required to obtain an ultrasound image.
- Define pulse repetition frequency and state the role it plays in frame rate.
- Name the main components of an ultrasound imaging system.
- Explain the preprocessing required to obtain digital data from ultrasound signals.
- Define the term *time gain compensation*.
- Explain how postprocessing is used to change the appearance of an ultrasound image.
- Describe several common artifacts in ultrasound images.
- List several common ultrasound quality control tests.



MARGIN FIGURE 21-1

A-mode (amplitude mode) of ultrasound display. An oscilloscope display records the amplitude of echoes as a function of time or depth. Points A, B, and C in the patient appear as peaks-A, B, and C in the A-mode display.

For each centimeter of depth, ultrasound travels 2 cm, 1 cm “out” for the transmitted pulse and 1 cm “back” for the reflected pulse. With the assumption of a speed of sound of 1540 m/sec for ultrasound in soft tissue, the time that corresponds to each centimeter of “depth” is

$$\frac{0.02 \text{ m}}{1540 \text{ m/sec}} = 0.000013 \text{ sec} = 13 \mu\text{sec}$$

The result is a conversion factor of 13 $\mu\text{sec/cm}$ for diagnostic ultrasound that is used to convert time to distance and vice versa.

This conversion factor is appropriate for reflection imaging in soft tissue where a speed of sound of 1540 m/sec is assumed to be reasonably accurate.

A-mode displays of a transcranial scan (echoencephalography) were used until the late 1970s to detect shifts of the midline in the brain of newborns with suspected hydrocephaly. This technique has been replaced by B-mode real-time scanning, computed tomography and magnetic resonance imaging.

PRESENTATION MODES

A-Mode

In the A-mode presentation of ultrasound images, echoes returning from the body are displayed as signals on an oscilloscope. The oscilloscope presents a graph of voltage representing echo amplitude (hence the term “A-mode”) on the ordinate, or y-axis, as a function of time on the abscissa, or x-axis. With the assumption of a constant speed of sound, time on the x-axis can be presented as distance from the ultrasound transducers (Margin Figure 21-1).

A-mode reveals the location of echo-producing structures only in the direction of the ultrasound beam. It has been used in the past to localize echo-producing interfaces such as midline structures in the brain (echoencephalography) and structures to be imaged in B-mode. A-mode displays are not found on most imaging systems used today. The concept of A-mode is however, useful in explaining how pixels are obtained from scan lines in B-mode imaging.

B-Mode

In B-mode presentation of pulse echo images (Margin Figure 21-2) the location of echo-producing interfaces is displayed in two dimensions (x and y) on a video screen. The amplitude of each echo is represented by the brightness value at the xy location. High-amplitude echoes can be presented as either high brightness or low brightness to provide either “white-on-black” or “black-on-white” presentations. Most images are viewed as white on black, so regions in the patient that are more echogenic correspond to regions in the image that are brighter (hence B for “brightness” mode).

B-mode images may be displayed as either “static” or “real-time” images. In static imaging the image is compiled as the sound beam is scanned across the patient, and the image presents a “snapshot” averaged over the time required to sweep the sound beam. In real-time imaging, the image is also built up as the sound beam scans across the patient, but the scanning is performed automatically and quickly, and one image follows another in quick succession. At image frequencies greater than approximately 24 per second, the motion of moving structures seems continuous, even though it may appear that the images are flickering (i.e., being flashed on and off). Images that are refreshed at frequencies greater than approximately 48 per second are free of flicker.

Real-time B-mode images are useful in the display of moving structures such as heart valves. They also permit the technologist to scan through the anatomy to locate structures of interest. Certain applications such as sequential slice imaging of organs are performed better with static imaging.

M-Mode

The M-mode presentation of ultrasound images is designed specifically to depict moving structures. In an M-mode display, the position of each echo-producing interface is presented as a function of time. The most frequent application of M-mode scanning is echocardiography, where the motion of various interfaces in the heart is depicted graphically on a cathode-ray tube (CRT) display or chart recording.

In a typical M-mode display (Margin Figure 21-3), the depths of the structures of interest are portrayed as a series of dots in the vertical direction on the CRT, with the position of the transducer represented by the top of the display. With the transducer in a fixed position, a sweep voltage is applied to the CRT deflection plates to cause the dots to sweep at a controlled rate across the CRT screen. For stationary structures, the dots form horizontal lines in the image. Structures that move in a direction parallel to the ultrasound beam produce vertical fluctuations in the horizontal trace to reveal their motion. The image may be displayed on a short-persistence CRT or storage scope and may be recorded on film or a chart recorder. Modern systems digitize the information and display it on a digital monitor.

TIME REQUIRED TO OBTAIN IMAGES

Image formation with ultrasound requires that echo information be received along discrete paths called “scan lines.” The time required to complete each line is determined by the speed of sound. If all of the information must be received from one line before another can be initiated, a fundamental limit is imposed on the rate at which ultrasound images can be acquired (Margin Figure 21-4).

Each line of information is obtained as follows. First, an ultrasound pulse of several nanoseconds [called the pulse time (PT)] is sent. The transducer is then quiescent for the remainder of the pulse repetition period (PRP), defined as the time from the beginning of one pulse to the next. During the quiescent time, echoes returning from interfaces within the patient excite the transducer and cause voltage pulses to be transmitted to the imaging device. These “echoes” are processed by the device and added to the image only if they fall within a preselected “listen time.” Acquisition of echoes during the listen time provides information about reflecting interfaces along a single path in the object—that is, the scan line (Margin Figure 21-5).

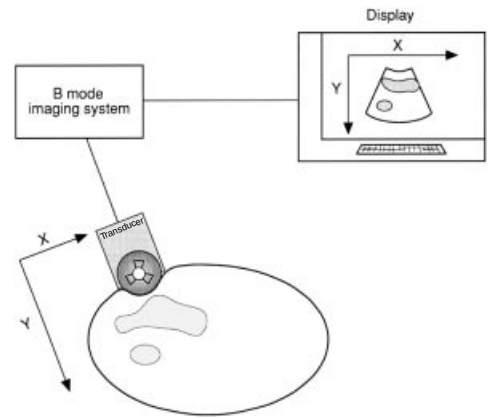
The PRP determines the maximum field of view (FOV), also known as depth of view (DOV)—that is, the length of the scan lines.

$$\text{PRP (sec)} = \text{FOV (cm)} \times 13 \times 10^{-6} \text{ sec/cm} \quad (21-1)$$

Example 21-1

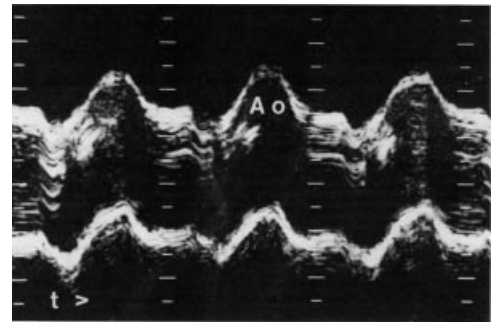
Find the maximum length of a scan line from an ultrasound unit having a PRP of 195 μs .

$$\begin{aligned} \text{Length of scan line} &= \frac{\text{PRP } (\mu\text{sec})}{13 \mu\text{sec/cm}} \\ &= \frac{195 \mu\text{sec}}{13 \mu\text{sec/cm}} \\ &= 15 \text{ cm} \end{aligned}$$



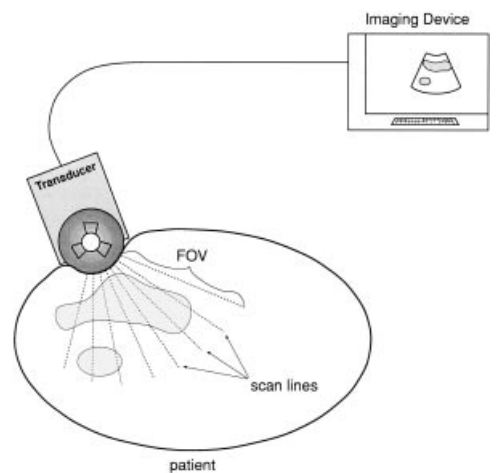
MARGIN FIGURE 21-2

B-mode, or brightness mode, ultrasound display. The amplitude of reflected signals is displayed as brightness at points of an image defined by their x - and y -coordinates.



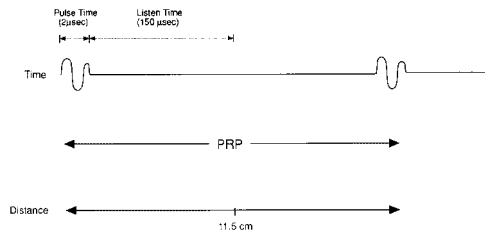
MARGIN FIGURE 21-3

A typical M-mode echocardiographic tracing.



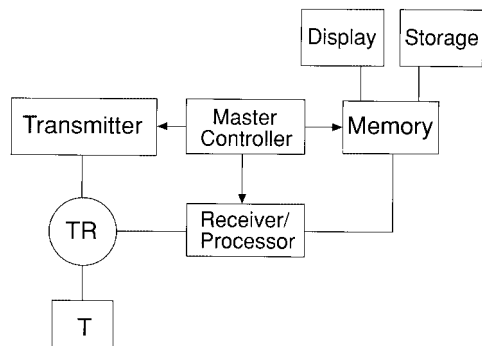
MARGIN FIGURE 21-4

A B-mode image consists of scan lines. The length of the scan lines determines the depth within the patient that is imaged [the field of view (FOV)].

**MARGIN FIGURE 21-5**

Ultrasonic echoes from a single scan line in A- or B-mode imaging. In the example shown here, a 150- μsec “listen time” corresponds to a depth [field of view (FOV)] of 11.5 cm. The pulse time (or pulse duration) is short when compared with the listen time. The time from the beginning of one pulse to the beginning of the next is the pulse repetition period (PRP).

Note that the PRP determines the *maximum* depth of view. The depth of view selected on the imaging device could be smaller than $\text{PRP}(\mu\text{s})/13(\mu\text{s}/\text{cm})\mu\text{sec}$ if the pulse listen time were shorter than the PRP.

**MARGIN FIGURE 21-6**

Main components of an ultrasound B-mode imaging system.

Ultrasound instrumentation for medical imaging was developed after World War II when personnel, who were familiar with military electronics (signal generators, amplifiers) and the use of sonar, returned to civilian life and wanted to transfer these technologies to medicine. One of the first experimental scanners was built in Denver, CO in 1949 by Joseph Holmes and Douglas Howry. It employed an amplifier from surplus Air Force radar equipment, a power supply from a record player, and a gun turret from a B-29 bomber.

The pulse repetition frequency (PRF) is the inverse of the PRP:

$$\text{PRF} = \frac{1}{\text{PRP}}$$

where the PRF has units of pulses per second, inverse seconds, or hertz. The concept of PRF is of special interest in pulsed Doppler ultrasound (Chapter 22).

The frame time (FT) is the time required to obtain a complete image (frame) made up of multiple scan lines (N):

$$\text{FT} = \text{PRP} \times N \quad (21-2)$$

Obviously, the frame time can be shortened by obtaining two or more scan lines simultaneously. Some linear arrays are capable of providing “parallel” acquisition of image data. For the purpose of problems presented here, we will assume that scan lines are acquired sequentially (i.e., one after the other).

A quantity such as frame time is a “period.” It is the time required to complete a periodic task. For any such quantity, there is the inverse, called a “frequency” or “rate,” that depicts the number of such tasks that can be performed per unit time. So the rate at which images may be acquired, the frame rate (FR), is the inverse of the frame time.

$$\text{FR} = \frac{1}{\text{FT}}$$

By substituting from Eqs. (21-1) and (21-2), we obtain

$$\text{FR} = \frac{1}{13 \times 10^{-6} \times \text{FOV} \times N} \quad (21-3)$$

The frame rate has units of frames per second, inverse seconds, or hertz.

Example 21-2

For an ultrasound image with a 10-cm field of view (FOV) and 120 lines of sight, find the minimum pulse repetition period (PRP), the minimum frame time (FT), and the maximum frame rate (FR).

$$\begin{aligned} \text{PRP} &= \text{FOV} \times 13 \times 10^{-6} \\ &= 10 \text{ cm} \times 13 \times 10^{-6} \text{ sec/cm} \\ &= 130 \times 10^{-6} \\ &= 130 \mu\text{sec} \end{aligned}$$

$$\begin{aligned} \text{FT} &= \text{PRP} \times N \\ &= 130 \times 10^{-6} \text{ sec} \times 120 \\ &= 15.6 \times 10^{-3} \text{ sec} \\ &= 15.6 \text{ msec} \end{aligned}$$

$$\begin{aligned} \text{FR} &= \frac{1}{\text{FT}} \\ &= \frac{1}{15.6 \times 10^{-3} \text{ sec}} \\ &= 64 \text{ sec}^{-1} \\ &= 64 \text{ frames/sec} \\ &= 64 \text{ Hz} \end{aligned}$$

An inverse relationship exists between the frame rate and each of the variables pulse repetition period, field of view, and number of scan lines. An increase in any of these variables decreases the frame rate in direct proportion.

SYSTEM COMPONENTS

The main components of an ultrasound B-mode imaging system are shown in Margin Figure 21-6. During a scan, the transmitter sends voltage pulses to the transducer. The pulses range from tens to hundreds of volts in amplitude. Returning echoes from the patient produce voltage pulses that are a few microvolts to a volt or so in amplitude. These small signals are sent to the receiver/processor for preprocessing functions such as demodulation, thresholding, amplification, time gain compensation, and detection, all described below. From the receiver/processor, signals are stored as digital values in random-access computer memory.

An automatic transmit/receive switch is provided to prevent large transmitter signals from finding their way into the sensitive receiver/processor. Failure of this switch could result in damage to the receiver/processor, as well as yield no image or one of diminished brightness. The latter condition would call for an increased gain setting beyond that normally used.

The functions of the components shown in Margin Figure 21-6 are coordinated by the master controller. The master controller provides reference signals against which the various components can time the arrival of echoes or keep track of which scan line is being processed.

For static B-mode imaging, a position-sensing device is added to Margin Figure 21-6. The device is attached to the transducer assembly to monitor the position and orientation of the transducer. Electronic components called “potentiometers” are used in the position-sensing device. When the potentiometer is turned, its resistance changes, producing a change in voltage in a position-sensing circuit. As the position-sensing device is moved, the potentiometers located at hinges provide voltages that are automatically converted to position coordinates (Margin Figure 21-7). For real-time B-mode imaging, the master controller encodes the echoes according to their location in space (Margin Figure 21-8). The echo signals are stored in memory in such a way that when read out as video signals, they appear on the display as brightness values in locations that are meaningful representations of their origins within the patient.

Ultrasound images are usually displayed on a CRT. M-mode data are usually recorded on a strip chart recorder. Static B-mode images usually are recorded on film with a hard-copy camera, and real-time B-mode images usually are preserved on videotape or disk.

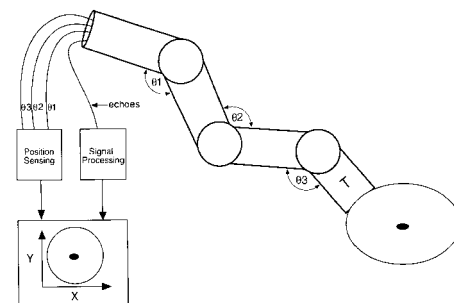
SIGNAL PROCESSING

Returning echoes contain a tremendous amount of information about the patient. As in all imaging modalities, methods used to process the echoes determine the information content of the final image. In B-mode imaging, each location in the image is associated with one value of echo amplitude. This value is recorded in memory and translated into brightness on the display screen. Deriving a single value from each echo is the goal of signal processing.

Signal processing may be divided into two categories. If the processing schemes to be used are determined prior to scanning the patient, they are termed *preprocessing schemes*. Postprocessing refers to signal or image processing that alters stored values of data after they have been acquired, (i.e., after the patient has been scanned).

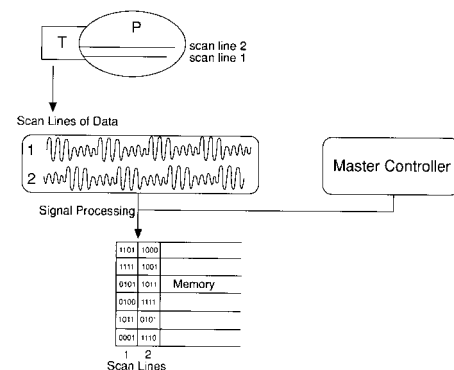
Preprocessing

The amplitude of an ultrasound echo is governed by the acoustic impedance mismatch at the interface where the echoes originated, the attenuation of intervening tissues, and the amplitude of the ultrasonic pulse that is sent out from the transducer. It is the presence and degree of acoustic impedance mismatch at interfaces that we wish to image. The attenuation by intervening tissues is usually considered an undesirable factor because it produces falloff of signal intensity with depth without yielding any useful information. This attenuation can be compensated for in the image by use of the



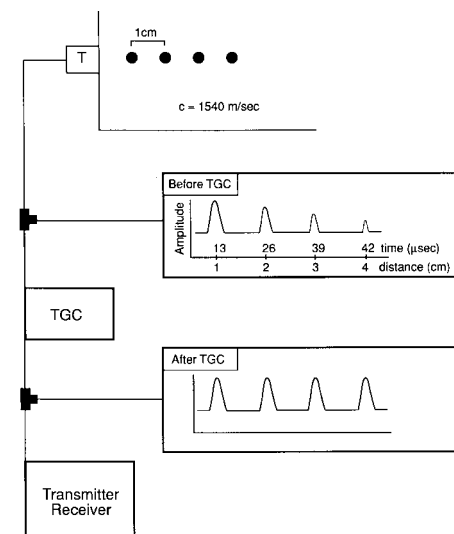
MARGIN FIGURE 21-7

The XY position—sensing circuits of a static B-mode scanner. Electric signals from potentiometers 1, 2, and 3 are proportional to the angles θ_1 , θ_2 , and θ_3 at the hinges of the articulated arm.



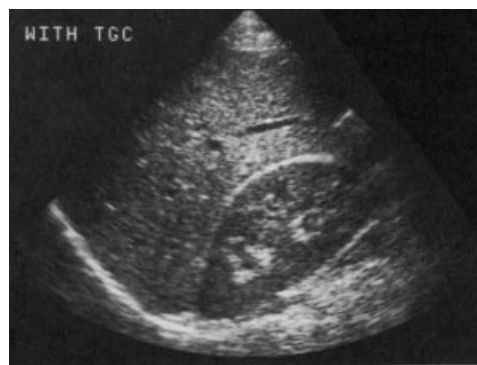
MARGIN FIGURE 21-8

In a real-time B-mode imaging system, spatial encoding of the echo signals is accomplished by storing scan-line data in memory in a methodical fashion.



MARGIN FIGURE 21-9

Illustration of the time gain compensation (TGC) principle. Identical steel rods are imaged in a phantom containing an attenuating medium. The time at which echoes return to the transducer (T) depends only upon the speed of sound in the medium. TGC compensates for attenuation by varying the amplification as a function of time of echo reception.



(a)



(b)

MARGIN FIGURE 21-10

Real-time B-mode images obtained with (A) and without (B) a properly adjusted TGC.

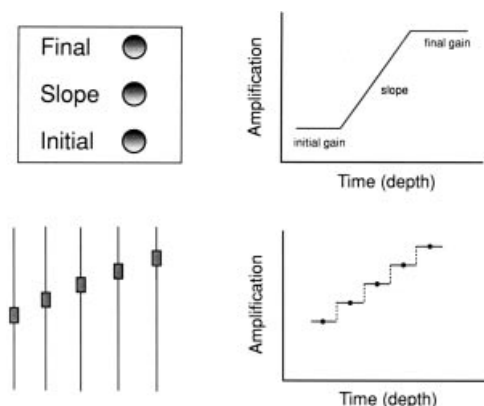
time gain compensation (TGC) circuit, sometimes called the depth gain compensation (DGC) or time-varied gain (TVG) circuit.

The action of the TGC circuit is shown in Margin Figure 21-9. A plot of uncorrected or “raw” amplitudes or echoes (in the form of an A mode display) from a succession of equivalent reflectors in a homogeneous medium shows a predictable decrease in signal amplitude with depth. The key to compensating for this falloff is the relationship between the depth and time of arrival of the echo at the transducer. Echoes that return at longer times (and therefore from greater depth) receive greater amplification to compensate for increased attenuation by intervening soft tissue. The curve that describes the amplification or gain required to compensate for the attenuation is called the TGC curve. Margin Figure 21-10 shows images obtained with and without a properly adjusted TGC.

The attenuation of different body parts and variations from one patient to another render a single factory setting of the TGC impractical. The TGC must be adjusted for each patient and readjusted during the scan as different tissues are encountered. In practice, most ultrasound systems do not allow specification of a continuous curve to describe the TGC. Instead, a few controls allow reasonable flexibility in determining the curve. Two common types of TGC controls are (1) a series of linear potentiometers for discrete setting of time delays at various depths and (2) a three-knob control that allows adjustment of initial gain, slope, and far gain. The two types of TGC systems are illustrated in Margin Figure 21-11.

Other types of signal preprocessing commonly performed in ultrasound imaging include rejection, rectification, enveloping, and classification (Margin Figure 21-12). Rejection eliminates signals that are too large or too small to be of diagnostic value. All remaining signals are rectified and enveloped, a process sometimes referred to as “low-pass” filtering. Echo intensity may then be defined in a number of ways. Three typical methods are to classify echoes according to (1) the height of the peak value, (2) the area under the peak, or (3) the maximum rate of rise or slope of the echo.

A final preprocessing step involves assigning a value in computer memory that corresponds to signal intensity. Margin Figure 21-13 shows a preprocessing “map” that demonstrates how signal levels measured in decibels are converted into arbitrary numbers to be stored in memory. If the map is linear, for example, then a 10% increase in signal level would be stored as a 10% larger value in memory. By using a nonlinear map, it is possible to accentuate part of the range of echoes at the expense of another. In Margin Figure 21-13 the larger signals, from about 15 to 30 dB, are mapped into stored values of 50 to 256, while smaller signals (0 to 15 dB) are mapped into values of 0 to 50. Thus, the upper half of the range of echoes is accentuated because it takes up four-fifths of the stored values. Relationships among the stronger signals will be accentuated because the gray levels in the display will be taken up disproportionately by the stronger signals. Smaller signals will all appear very dark, with little differentiation. Therefore, the map illustrated in Margin Figure 21-13 provides high signal separation.

**MARGIN FIGURE 21-11**

Three-knob TGC control (A) and a linear potentiometer TGC control (B) along with the TGC curves produced.

Postprocessing

The value stored for each location in the image (each picture element, or pixel) is eventually displayed on a monitor as a level of brightness. That is, the echo intensities are displayed as brightness values varying from black to white. As with preprocessing, one may choose a “map” for postprocessing that may be either linear or nonlinear. A nonlinear map can be designed to emphasize some parts of the range of stored values. In Margin Figure 21-14 a postprocessing map is shown that maps the lower stored values into most of the gray scale, thereby emphasizing differences among the lower values. Note that if the image were preprocessed with the map shown in Margin Figure 21-13 and then postprocessed with the map in Margin Figure 21-14, the result would be the same as if a linear map were used for both. That is, postprocessing can either “undo” or enhance the effects of preprocessing.

Other types of postprocessing features that may be present in an ultrasound device include zoom, region of interest, and digital filtering (see Chapter 10).

DYNAMIC RANGE

The ratio of the largest to the smallest echoes processed by components of an ultrasound device is known as the dynamic range of the device. In general, the dynamic range decreases as signals pass through the imaging system (Margin Figure 21-15) because operations such as TGC and rejection eliminate small and large signals. Echoes returning from tissues can have dynamic ranges of 100 to 150 dB. The dynamic range in decibels may be easily converted to a ratio of amplitudes or intensities.

Example 21-3

Find the amplitude and intensity ratios corresponding to a dynamic range of 100 dB. For amplitudes,

$$\text{dB} = 20 \log \frac{A}{A_0}$$

can be rewritten as

$$\frac{\text{dB}}{20} = \log \frac{A}{A_0}$$

or

$$10^{\text{dB}/20} = \frac{A}{A_0}$$

So, for 100 dB we have

$$10^5 = 100,000 = \frac{A}{A_0}$$

The strongest echoes have 100,000 times the amplitude of the weakest echoes. Similarly, for intensity,

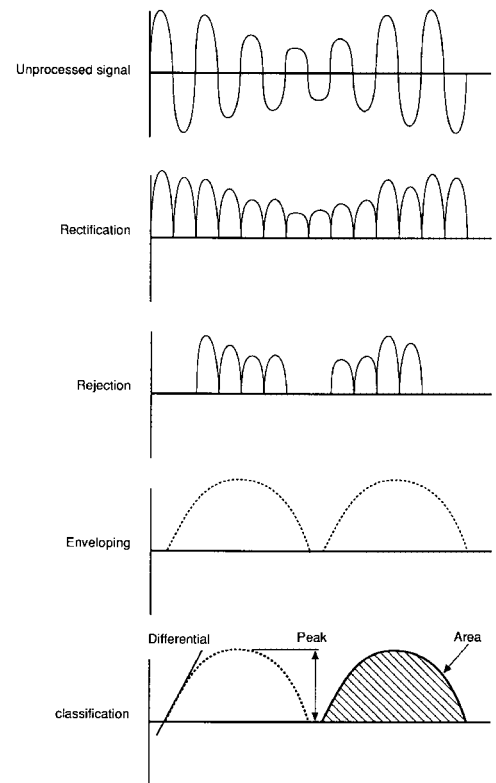
$$\text{dB} = 10 \log \frac{I}{I_0}$$

so that

$$10^{\text{dB}/10} = 10^{10} = 10,000,000,000 = \frac{I}{I_0}$$

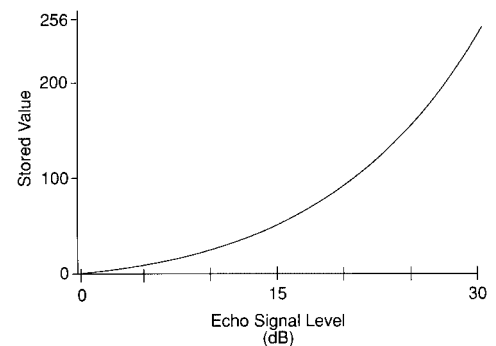
The strongest echoes have 10^{10} , or 10 billion, times the intensity of the weakest echoes.

Some of the dynamic range of echoes is caused by attenuation as the signals traverse several centimeters of tissue. The TGC circuit amplifies the weak echoes that arrive later, and rejection circuits eliminate low-amplitude noise. In this manner, the dynamic range is reduced, or “compressed,” to about 40 dB in the receiver/processor. This dynamic range may be retained in memory while a range of about 20 dB is displayed on the system monitor.

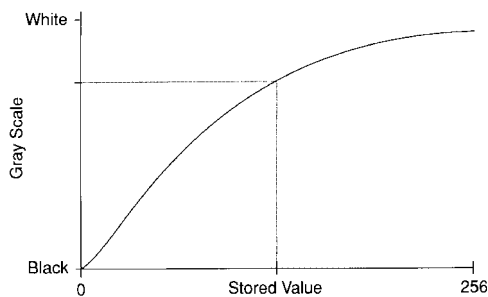


MARGIN FIGURE 21-12
Signal preprocessing in ultrasound.

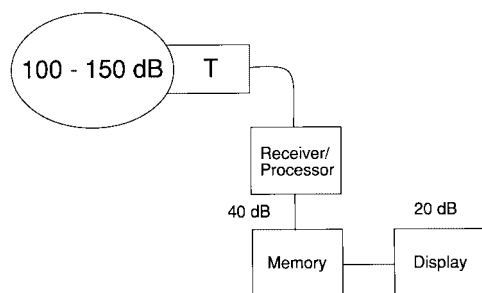
The terms “enveloping” and “low-pass filtering” are used interchangeably in signal processing. A low-pass filter removes higher frequencies but allows lower ones to pass through the circuit. If a complex signal (one consisting of varying amounts of many different frequencies) is sent through a low-pass filter, the resulting signal will look something like the outline or envelope of the original signal.



MARGIN FIGURE 21-13
Preprocessing “map.” The graph shows how echo signal levels are converted to stored values. This particular example shows “high signal separation,” with the upper half of the amplitude range taking up the upper 4/5 of the stored values.

**MARGIN FIGURE 21-14**

Postprocessing “map.” The graph shows how stored values from memory are converted to gray-scale values. This particular example shows “low signal separation,” with the lower half of the stored values taking up the lower 4/5 of the gray scale (brightness values vary from black to white).

**MARGIN FIGURE 21-15**

Dynamic range of signals at different stages of the ultrasound imaging system.

**MARGIN FIGURE 21-16**

Acoustic shadowing in an ultrasound image. Stones in a gallbladder reduce the transmission of sound. The display records lower amplitude echoes or “shadows” beyond the region of the stones.

■ ULTRASOUND IMAGE ARTIFACTS

Even though ultrasound systems produce digital images, it is impossible to identify pixel values in typical clinical images with specific fundamental properties of tissue in the way that x-ray CT numbers are associated with linear attenuation coefficients. Although pixel values in ultrasound images are influenced by fundamental properties such as acoustic impedance and physical density, each pixel value is determined by interrelationships among surrounding materials that are too complex to unfold quantitatively. However, these interrelationships sometimes create artifacts that are recognizable in a qualitative sense.

Speckle

The physical principles behind the phenomenon of speckle were described in Chapter 20. Although the presence of speckle in an image indicates the presence of features in the patient that are smaller than the wavelength of the ultrasound, there is no one-to-one correspondence between a dot in a speckle pattern and the location of a feature. The speckle pattern changes from frame to frame, even for stationary objects, and therefore is not a simple indicator of even the number of such objects. However, various attempts have been made to determine the presence or absence of disease in organs from statistical analysis of speckle patterns.¹

Shadowing and Enhancement

If some object within the patient has a larger attenuation coefficient than the material that lies beyond it, then the settings of the TGC circuit that would provide appropriate compensation for normal tissue will undercompensate and cause the region beyond the object to appear less echogenic. This phenomenon is referred to as acoustic shadowing (Margin Figure 21-16). Similarly, if the object in the path of the ultrasound beam has a lower attenuation coefficient than its surroundings, acoustic enhancement may result.

Multiple Pathway

Various types of multiple-pathway artifacts occur in ultrasound images (see Chapter 20). When an echo returns to a transducer, the imaging device assumes that the sound traveled in a straight line following a single reflection from some interface in the patient. The scan converter then places the brightness value at an appropriate location in the image. If the actual path of the echo involved multiple reflections, the echo would take longer to return, and the scan converter would place the interface at a greater depth in the image.

Refraction

Refraction sometimes causes displacement of the sound beam as it crosses tissue boundaries. Because the scan converter assumes that the ultrasound travels in straight lines, refraction causes displacement errors in positioning reflective interfaces in the image. A refraction artifact is illustrated in Margin Figure 21-17.

■ QUALITY CONTROL

Frequent testing of the performance of the ultrasound imaging system by using phantoms whose properties are well understood can uncover problems that may be too subtle to discern in clinical images.² Standard phantoms are shown in Margin

Figure 21-18. Their properties and some protocols for typical performance tests are discussed below.

Shown in Margin Figure 21-18A is an example of a high-contrast test object for evaluation of ultrasound imaging systems. The test object mimics the speed of sound in soft tissue and contains a mixture of ethanol and water or NaCl and water. With this test object, the depth-measuring properties of ultrasound systems can be evaluated. "An equivalent speed of sound" also implies that the wavelength c/ν , near zone length r^2/λ for unfocused transducers, and the focal length for focused transducers are the same in the test medium as in soft tissue. Therefore, this phantom may be used to evaluate axial resolution at an appropriate depth.

The standard American Institute of Ultrasound in Medicine (AIUM) 100-mm test object (Margin Figure 21-18A) is an example of a high-contrast test object that employs a medium with a speed of sound equal to that of soft tissue. Reflected signals are produced by rods having diameters of 0.75 mm located at depths specified to an accuracy of ± 0.25 mm. The diameter and accuracy of location of the rods exceed the specifications for minimum resolution and distance for typical imaging systems.

A tissue-equivalent phantom (Margin Figure 21-18B) mimics more of the ultrasonic properties of human soft tissue as compared with the high-contrast test object. These properties may include scatter characteristics at typical frequencies as well as overall attenuation. The phantom provides an image with a texture pattern similar to that of liver parenchyma with a few low-contrast "cysts" in the phantom.

Quality Control Tests

Dead Zone

A small portion of the image at the transducer-tissue interface is usually saturated with echoes because of the spatial pulse length of the transducer and multiple reverberations from the transducer-tissue interface. Examination of a group of high-contrast rods located near the surface of the phantom reveals the extent of this problem. The minimum depth at which a high-contrast rod can be resolved is usually quoted as the depth of the dead zone.

Axial Resolution

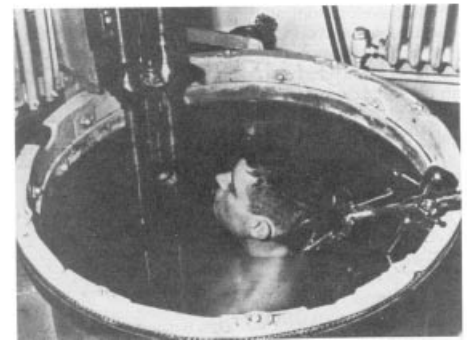
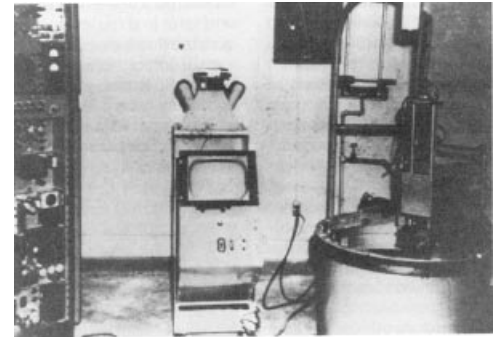
Any equipment problem that reduces the bandwidth or increases the spatial pulse length of the system interferes with axial resolution. Examples of such problems include cracks in the crystal materials and separation of the facing or backing material. Axial resolution is measured by attempting to resolve (produce a clear image of) rods that are closely spaced along the axis of the sound beam.

Lateral Resolution

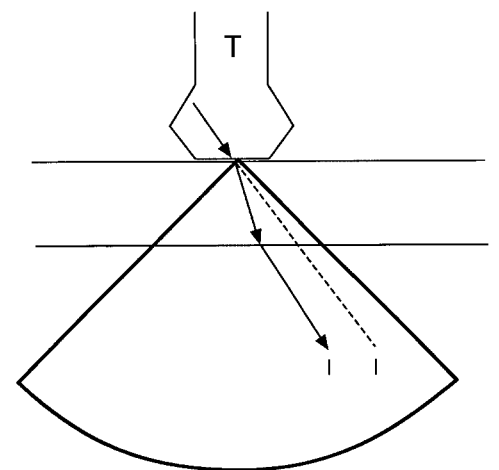
There are several methods to evaluate lateral resolution, defined as resolution in a direction perpendicular to the axis of the ultrasound beam. A scan of rod groups that form a line parallel to the scanning surface can be used. Alternatively, rods that are perpendicular to the axis of the beam may be scanned. Because the lateral resolution is the width of the ultrasound beam, the apparent width of the rods scanned in this fashion (Margin Figure 21-19) yields the resolution.

Distance Accuracy Measurement

Distance measurements play a key role in ultrasound dating of fetuses. Errors in distance calibration may lead to a change in the gestational age estimate that would adversely affect patient management. To perform the distance accuracy measurement, high-contrast rods with a separation of at least 10 cm are scanned. Distance

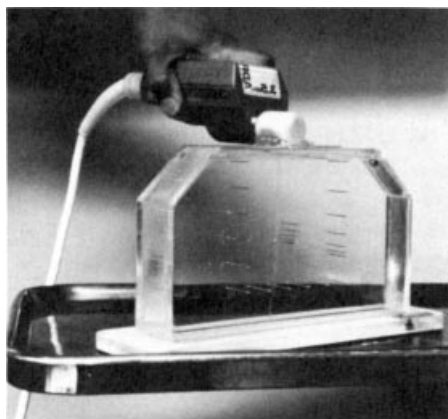


One of the first ultrasonic scanners for imaging the human body required that the patient be immersed in warm saline solution in a large metal tank in which a transducer rotated about the patient. Head scanning was not often performed with this technique. Shown here is the compound scanner (Somascope) developed by Howry and co-workers from a B-29 gun turret. (From *Am. J. Digest Dis.* 1963; 8:12. Harper and Row, Hoeber Medical Division. Used with permission.)

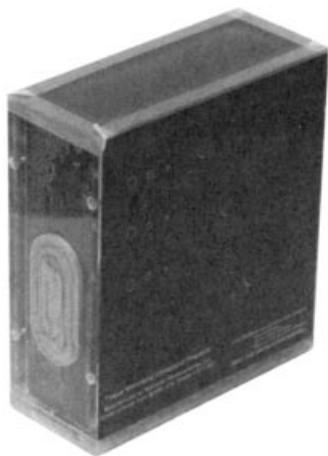


MARGIN FIGURE 21-17

Refraction artifact. The object on the left appears shifted to the right and at a slightly greater depth because of refraction in a layer having lower speed of sound than the rest of the tissue being imaged.



(a)



(b)

MARGIN FIGURE 21-18

Standard ultrasound phantoms. AIUM 100-mm test object (A) and a tissue-equivalent phantom (B).

measurement options that are available on the imaging system (marker dots, calipers, etc.) are then used to determine the distance from the image. Because the rod spacings in the test object or phantom are known, the distance error may be reported as a percentage of the true value. For example, a measured distance of 10.5 for a known distance of 10.0 cm is reported as a 5% error. A reasonable goal for most imaging systems currently in use is an error of 2% or less measured over 10 cm.

Sensitivity and Uniformity

There are a number of ways to gauge the ability of an ultrasound system to record faint sources of echoes (sensitivity) and provide a consistent response to echoes (uniformity).³ Rod groups both perpendicular and parallel to the scanning surface may be scanned, and the brightness and size of each rod may be compared. If a rod group perpendicular to the scanning surface is scanned in an attenuating phantom, then the TGC settings required to render a uniform rod image appearance may be recorded. Images obtained with those TGC settings may be reviewed periodically as a constancy check.

Display and Printing

The consistency of the recording medium may be evaluated with test patterns supplied by the manufacturer. Basically, the appearance of the “hard copy” should be the same as its appearance on the system monitor. This parameter should be measured qualitatively on all systems. More quantitative evaluations involving measurement of monitor luminance are possible⁴ but not routine. If a test pattern is not available in vendor-supplied software, it is possible to use any image, such as a typical image of the tissue-equivalent phantom, to establish photographic consistency.

It is most important to separate performance of the image acquisition system from performance of the recording device. This is best accomplished by using a test pattern that is generated independently from the image acquisition system—that is, an “electronic” test pattern. An electronic test pattern that is frequently used for this purpose is a pattern that has been specified by the Society of Motion Picture and Television Engineers (SMPTE). The SMPTE pattern is shown in Margin Figure 21-20.

Frequency of Testing

The quality control tests outlined in this section should be performed monthly and after any service call that may affect the parameters measured.⁵ More extensive tests may be performed by qualified personnel upon delivery of the unit.

PROBLEMS

- | | |
|--|--|
| <p>21-1. If A, B, and M mode ultrasound display modes are described as X versus Y, what are the X and Y scales for each mode?</p> <p>*21-2. How many scan lines may be obtained during a scan of a patient when the depth of view (DOV) is 8 cm if the total scan time is 13.4 msec? Assume sequential acquisition of scan lines.</p> <p>*21-3. If the pulse repetition period is doubled, the field of view cut in half, and the number of scan lines doubled what is the effect on</p> | <p>maximum allowable frame rate? Assume sequential acquisition of scan lines.</p> <p>21-4. Describe the function of the master controller of a real-time ultrasound B-mode imaging system.</p> <p>*21-5. What phenomenon is compensated for by the time gain compensation (TGC) circuit?</p> |
|--|--|

*For those problems marked with an asterisk, answers are provided on p. 493.

SUMMARY

- Display modes include
 - Amplitude
 - Brightness
 - Motion
- Frame rate is inversely proportional to depth of view, number of scan lines, and pulse repetition frequency.
- The main components of an ultrasound imaging system are
 - Transducer
 - Transmit/receive switch
 - Transmitter
 - Receiver/processor
 - Master controller
 - Memory
 - Display
 - Storage
- The time gain compensation system varies the amplification of return signals from each scan line in a variable manner as a function of time after the sent pulse.
- Preprocessing steps include
 - Rectification
 - Pulse height rejection
 - Enveloping (low-pass filtering)
 - Classification (e.g., differential, peak, area, alone or some combination)
- In preprocessing, echo signals are assigned to stored values according to according to a functional form or curve.
- In postprocessing, stored values are assigned to the gray scale according to a functional form or curve.
- Artifacts include
 - Speckle
 - Acoustic shadowing
 - Multiple pathway
 - Refraction
- Ultrasound quality control tests include
 - Dead zone
 - Axial resolution
 - Lateral resolution
 - Distance accuracy
 - Sensitivity and uniformity

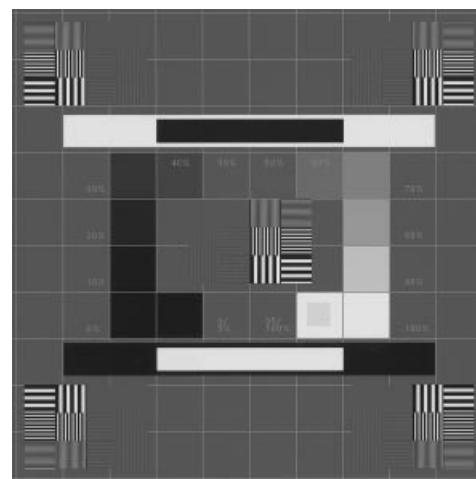


(a)

(b)

MARGIN FIGURE 21-19

Scan of a rod group in an ultrasound phantom demonstrates beam width (lateral resolution).



MARGIN FIGURE 21-20

The test pattern promulgated by the Society of Motion Pictures and Television Engineers (SMPTE), known colloquially as the "simtē pattern." This pattern, which tests image quality parameters such as high and low contrast resolution, distortion, and linearity, may be used as an electronic data file to test the fidelity of displays, hard-copy devices, or storage and transmission systems. (ref: Society of Motion Pictures and Television Engineers (SMPTE). Specifications for Medical Diagnostic Imaging Test Pattern for Television Monitors and Hard-Copy Recording Cameras. Recommended Practice RP 133-1986. *SMPTE J.* 1986; **95**:693–695.)

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DOPPLER EFFECT

OBJECTIVES 344

ORIGIN OF DOPPLER SHIFT 344

Pulsed Doppler 348

Spectral Analysis 349

Frequency Spectrum 349

Spectral Trace 349

Color Mapping 350

Quantitative Use of Doppler 350

LIMITATIONS OF DOPPLER SYSTEMS 351

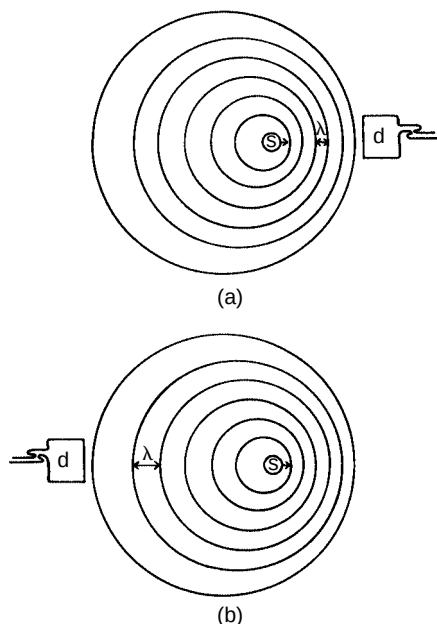
Pulse Repetition Frequency 351

Spectral Broadening 352

PROBLEMS 352

SUMMARY 352

REFERENCES 353

**MARGIN FIGURE 22-1**

Principles of Doppler ultrasound. **A:** Source moving toward a stationary detector **d**. **B:** Source moving away from a stationary detector **d**.

When an emergency vehicle with its siren on passes someone on the street, a shift in frequency of one octave on the chromatic musical scale corresponds to a velocity difference between the siren and the listener of 40 miles per hour, two octaves corresponds to 80 miles per hour, and so on.

OBJECTIVES

After completing this chapter, the reader should be able to:

- Write the Doppler equation and use it to calculate the Doppler shift.
- Describe the effect of Doppler angle upon frequency shift and describe how error in estimation of the angle influences the accuracy of velocity estimation.
- Determine volumetric flow from average velocity and vessel cross-sectional area.
- Name the main components of a Doppler ultrasound imager.
- Calculate range gate size and depth from timing information.
- Describe the differences between information obtained from pulsed-wave versus continuous-wave Doppler.
- Describe the main features of a frequency spectrum and describe how it is used in constructing the Doppler spectral trace.
- Describe how color is assigned in color Doppler.
- Discuss some reasons for spectral broadening.

Ultrasound is used not only for display of static patient anatomy but also for identification of moving structures in the body. Approaches to the identification of moving structures include real-time pulse-echo imaging, motion mode (M-mode) display of reflected ultrasound pulses, and the Doppler-shift method. Discussed in this chapter are the basic principles of the Doppler-shift method. The Doppler method has a number of applications in clinical medicine, including detection of fetal heartbeat, detection of air emboli, blood pressure monitoring, detection and characterization of blood flow, and localization of blood vessel occlusions.^{1,2}

ORIGIN OF DOPPLER SHIFT

When there is relative motion between a source and a detector of ultrasound, the frequency of the detected ultrasound differs from that emitted by the source. The shift in frequency is illustrated in Margin Figure 22-1. In Margin Figure 22-1A, an ultrasound source is moving with velocity v_s toward the detector. After time t following the production of any particular wave front, the distance between the wave front and the source is $(c - v_s)t$, where c is the velocity of ultrasound in the medium. The wavelength λ of the ultrasound in the direction of motion is shortened to

$$\lambda = \frac{c - v_s}{v_0}$$

where v_0 is the frequency of ultrasound from the source. With the shortened wavelength, the ultrasound reaches the detector with an increased frequency ν :

$$\begin{aligned} \nu &= \frac{c}{\lambda} = \frac{c}{(c - v_s)/v_0} \\ &= v_0 \left(\frac{c}{c - v_s} \right) \end{aligned}$$

That is, the frequency of the detected ultrasound shifts to a higher value when the ultrasound source is moving toward the detector. The shift in frequency $\Delta\nu$ is

$$\begin{aligned} \Delta\nu &= \nu - v_0 = v_0 \left(\frac{c}{c - v_s} \right) - v_0 \\ &= v_0 \left(\frac{v_s}{c - v_s} \right) \end{aligned}$$

If the velocity c of ultrasound in the medium is much greater than the velocity v_s of the ultrasound source, then $c - v_s \simeq c$ and

$$\Delta\nu = \nu_0 \left(\frac{v_s}{c} \right)$$

A similar expression is applicable to the case in which the ultrasound source is stationary and the detector is moving toward the source with velocity v_d . In this case the Doppler-shift frequency is approximately

$$\Delta\nu = \nu_0 \left(\frac{v_d}{c} \right)$$

where $c \gg v_d$.

If the ultrasound source is moving away from the detector (Margin Figure 22-1B), then the distance between the source and a wavefront is $ct + v_s t = (c + v_s)t$, where t is the time elapsed since the production of the wavefront. The wavelength λ of the ultrasound is

$$\lambda = \frac{c + v_s}{\nu_0}$$

and the apparent frequency ν is

$$\begin{aligned} \nu &= \frac{c}{\lambda} \\ &= \frac{c}{(c + v_s)/\nu_0} \\ &= \nu_0 \left(\frac{c}{c + v_s} \right) \end{aligned}$$

That is, the frequency shifts to a lower value when the ultrasound source is moving away from the detector. The shift in frequency $\Delta\nu$ is

$$\begin{aligned} \Delta\nu &= \nu - \nu_0 = \nu_0 \left(\frac{c}{c + v_s} \right) - \nu_0 \\ &= \nu_0 \left(\frac{-v_s}{c + v_s} \right) \end{aligned}$$

where the negative sign implies a reduction in frequency. If the velocity c of ultrasound is much greater than the velocity v_s of the source, $c + v_s \simeq c$, and $\Delta\nu = \nu_0(-v_s/c)$. A similar expression is applicable to the case where the ultrasound source is stationary and the detector is moving away from the source with velocity v_d :

$$\Delta\nu = \nu_0 \left(\frac{-v_d}{c} \right)$$

when $c \gg v_d$.

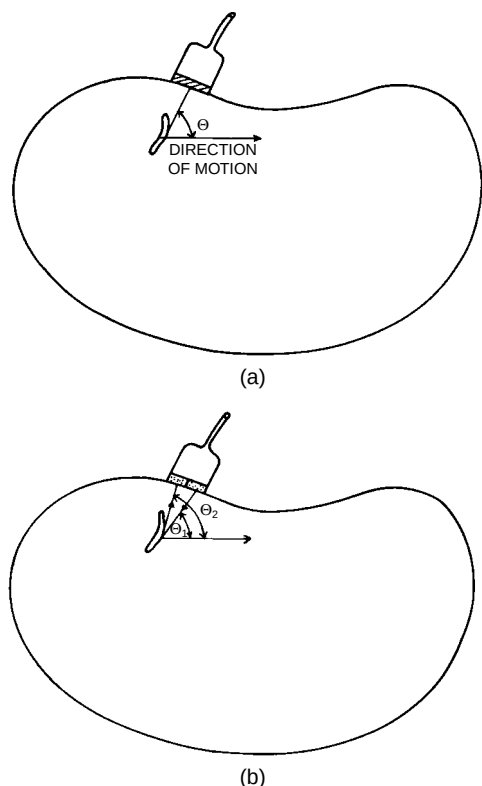
If the source and detector are at the same location and ultrasound is reflected from an object moving toward the location with velocity v , the object acts first as a moving detector as it receives the ultrasound signal and then as a moving source as it reflects the signal. As a result, the ultrasound signal received by the detector exhibits a frequency shift (when $c \gg v$)

$$\Delta\nu = 2\nu_0 \frac{v}{c}$$

Similarly, for an object moving away from the source and detector, the shift in frequency $\Delta\nu$ is

$$\Delta\nu = 2\nu_0 \left(\frac{-v}{c} \right)$$

where the negative sign indicates that the frequency of the detected ultrasound is lower than that emitted by the source.

**MARGIN FIGURE 22-2**

A: Angle θ between an incident ultrasound beam and the direction of motion of the object. **B:** For a dual probe with separate transmitting and receiving transducers, the angle θ is the average of the angles θ_1 and θ_2 that the transmitted and detected signals make with the direction of motion.

Austrian physicist Christian Doppler (1803–1858) first described the effect that bears his name in a paper published in the proceedings of the Royal Bohemian Society of Learning in 1843. The paper's title translates into English as: "Concerning the colored light of double stars." He explained the shift in color (frequency of light) of stars as their velocities changed relative to earth. While this effect is too small to result in a detectable color shift in measurements available at that time, he demonstrated the reality of this effect in 1845 using sound waves. Two trumpeters who could maintain a steady note with perfect pitch were recruited. One performed at the side of a railway, and the other performed on a moving train. The difference in pitch (frequency of sound) was apparent to listeners.

The discussion above has assumed that the ultrasound beam is parallel to the motion of the object. For the more general case where the ultrasound beam strikes a moving object at an angle θ (Margin Figure 22-2A), the shift in frequency $\Delta\nu$ is

$$\Delta\nu = 2\nu_0\left(\frac{v}{c}\right)\cos\theta \quad (22-1)$$

A negative sign in this expression would imply that the object is moving away from the ultrasound source and detector and that the frequency of detected ultrasound is shifted to a lower value. The angle θ is the angle between the ultrasound beam and the direction of motion of the object. If the ultrasound source and detector are displaced slightly (Margin Figure 22-2B), then θ is the average of the angles that the transmitted and reflected beams make with the motion of the object. For the small displacement between ultrasound transmitter and detector commonly used in Doppler ultrasound units, this assumption contributes an error of only 2% to 3% in the estimated frequency shift.³

Example 22-1

A 10-MHz ultrasound is used for the detection of blood flow. For blood flowing at 15 cm/sec toward the ultrasound source, find the Doppler-shift frequency. Assume that the angle θ is zero so that $\cos\theta = 1$.

$$\begin{aligned} \Delta\nu &= 2\nu_0\left(\frac{v}{c}\right)\cos\theta \\ &= (2)(10\text{ MHz})\left(\frac{15\text{ cm/sec}}{154,000\text{ cm/sec}}\right) \\ &= 1.9\text{ kHz} \end{aligned} \quad (22-1)$$

The audible frequency range is approximately 20 Hz to 20 kHz. Mechanical vibrations within this frequency range, if delivered with sufficient amplitude, are registered by the ear as audible sound. For physiologic flow rates the Doppler shift (the difference in frequency between sent and received signals) falls in this range and can be used to drive an audio speaker. The listener is not able to derive quantitative velocity information from the audible output but can appreciate changes in signal frequency and intensity. These changes may be caused by variations in the spatial location or angle of the probe with respect to the patient or by variations in the Doppler shift caused by temporal changes in flow within the patient during the cardiac cycle. As flow increases, the frequency of the sound (its "pitch") increases. The listener does not hear a pure single-frequency note but instead hears a complex assortment of frequencies similar to the sound of ocean waves breaking on the seashore. The assortment of frequencies, referred to as the spectral bandwidth of the Doppler signal, is discussed in detail later in this chapter.

The term involving $\cos\theta$ in the Doppler-shift equation [Eq. (22-1)] describes the variation in frequency shift as the sonation angle (the angle between the Doppler probe and the moving material) is varied. The shift is greatest when the longitudinal axis of the probe is oriented along the direction of motion. This occurs when the transmitted beam of sound is parallel (sonation angle = 0 degrees) or opposed to (sonation angle = 180 degrees) the direction of motion. Theoretically, no Doppler shift occurs when the probe is exactly perpendicular to the direction of motion. In practice, a small Doppler shift may be detected when the probe appears to be perpendicular to the direction of blood flow in a vessel. The small shift is due to several phenomena such as transverse components of flow in the vessel, detection of signals from side lobes, and so on.

The presence of the cosine term in the Doppler-shift equation complicates the direct measurement of velocity by the Doppler method. The only quantity that is measured directly by a typical Doppler ultrasound unit is Doppler shift. Determination of the velocity of blood in units such as centimeters per second requires an estimate

of the sonation angle. Errors in the estimate of this angle (using on-screen cursors) limit the accuracy of velocity estimates. This error is more significant at larger angles (near 90 degrees) than at smaller angles.

Example 22-2

A 10-MHz ultrasound probe detects a 1.4-kHz Doppler shift at a sonation angle estimated as 45 degrees. Find the estimated velocity of blood flow and the percent error in the estimate if the angle of sonation is incorrect by as much as 3 degrees:

$$\begin{aligned}\Delta v &= \frac{2v_0 v \cos \theta}{c} \\ v &= \frac{\Delta v c}{2v_0 \cos \theta} \\ &= \frac{(1.4 \times 10^3 \text{ sec}^{-1})(1.54 \times 10^5 \text{ cm/sec})}{2(10 \times 10^6 \text{ sec}^{-1})} \frac{1}{\cos \theta} \\ &= (10.8 \text{ cm/sec}) \frac{1}{\cos \theta}\end{aligned}$$

If $\theta = 45$ degrees

$$v = \frac{10.8}{0.707} = 15.3 \text{ cm/sec}$$

But if θ actually equaled 48 degrees instead of 45, then

$$v = \frac{10.8}{0.669} = 16.1 \text{ cm/sec}$$

The percent error is

$$\% \text{ error} = \frac{16.1 - 15.3}{15.3}(100) = 5.5\%$$

The error is slightly smaller if θ were actually 42 degrees.

Determining the volumetric flow of blood in units of cubic centimeters per second requires an estimate of the area of the vessel as well as the sonation angle. Furthermore, it assumes measurement of average quantities that are not always valid. Volumetric flow Q is the product of the average velocity v and the cross-sectional area A of the vessel:

$$Q = vA \quad (22-2)$$

In clinical measurements of volumetric flow, several sources of error reduce the accuracy below what would be expected based upon the intrinsic precision of Doppler-shift measurements. The measurement of vessel area is typically based upon measurement of vessel diameter in the image. However, the angle at which the image plane “cuts through” the vessel may cause either underestimation or overestimation of diameter. Also, the Doppler unit may not sample the average velocity in a vessel. Flow in a vessel (as in a mountain stream) does not have a uniform cross-section. It is usually greatest at the center and decreases to near zero at the vessel wall. Complex flow “profiles” are possible, particularly if a stenosis, bifurcation, or plaque formation is present. An accurate estimate of average velocity requires sampling of velocity at different radii within the vessel (at several points across the “stream”). It is possible to make some simplifying assumptions. For example, in laminar (nonturbulent) flow the average velocity is half the maximum value.⁴

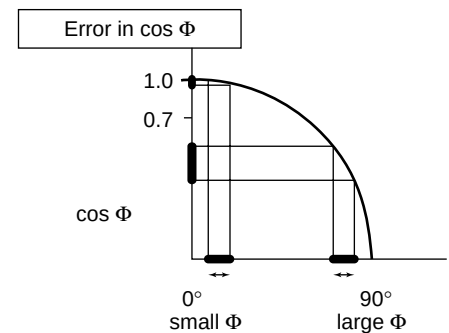
Example 22-3

Find the volumetric flow in a 3-mm-diameter vessel in which the average velocity (v) of blood is estimated to be 10 cm/sec.

An error in a sonographer's ability to estimate the Doppler angle in a real-time B-mode display affects the estimated value of blood velocity. The same error has a greater effect at angles near 90° than at angles near 0°. The error occurs because calculation of velocity from the Doppler frequency shift [Eq. (22-1)] involves multiplication by the cosine of the Doppler angle. This affect is appreciable even if the error is small.

If we assume an error of ± 3 degrees, then at an actual Doppler angle of 10° the sonographer's estimate will lead to a variation in the value of the cosine function from $\cos(7^\circ) = 0.993$ to $\cos(13^\circ) = 0.974$ where the true value of $\cos(10^\circ) = 0.985$. The error is less than $\pm 1\%$.

At an actual Doppler angle of 80° the same error of 3° leads to a variation in the value of the cosine function from $\cos(77^\circ) = 0.225$ to $\cos(83^\circ) = 0.122$ where the true value of $\cos(80^\circ) = 0.174$. Here, the error is approximately 30%!



MARGIN FIGURE 22-3

Because of the dependence of the Doppler shift upon the cosine of the Doppler angle, calculation of the velocity of an object from an observation of its Doppler shift requires estimation of the Doppler angle. This estimate (usually acquired through observation of a freeze frame of a real-time B mode ultrasound image) could be off by a few degrees. In the graph shown here we see that if the few-degree error (horizontal axis) occurs at a small Doppler angle (close to parallel to the direction of flow), the error introduced by the cosine function (vertical axis) is relatively small. If the same few degree error occurs at a larger Doppler angle (close to perpendicular to the direction of flow), then the error introduced by the cosine function is greater.

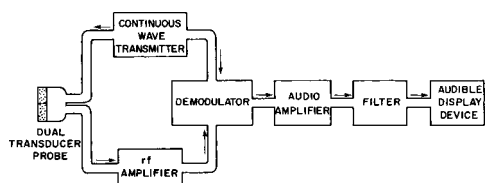
**MARGIN FIGURE 22-4**

Diagram of a continuous-wave Doppler unit.

First, find the cross-sectional area (A) of the vessel.

$$A = \pi \left(\frac{d}{2} \right)^2 = 3.14 \left(\frac{0.3 \text{ cm}}{2} \right)^2$$

$$= 0.07 \text{ cm}^2$$

The volumetric flow (Q) is

$$Q = vA$$

$$= (10 \text{ cm/sec})(0.07 \text{ cm}^2)$$

$$= 0.7 \text{ cm}^3/\text{sec} \quad (22-2)$$

A continuous-wave Doppler unit is diagramed in Margin Figure 22-4. The Doppler probe contains separate transmitting and receiving transducers. Application of a continuous electrical signal to the transmitting transducer produces a continuous ultrasound beam at a prescribed frequency. As the beam enters the patient, it is reflected from stationary and moving surfaces, and the reflected signals are returned to the receiver. Suppose that two signals are received, one directly from the transmitting transducer and the other from a moving surface. These signals interfere with each other to produce a net signal with a “beat frequency” equal to the Doppler-shift frequency. This Doppler-shift signal is amplified in the radio-frequency (RF) amplifier. The amplified signal is transmitted to the demodulator, where most of the signal is removed except the low-frequency beat signal. The beat signal is amplified in the audio amplifier and transmitted through the filter (where frequencies below the beat frequency are removed) to the loudspeaker or earphones for audible display. In practice, many moving interfaces reflect signals to the receiver, and many beat frequencies are produced.

Pulsed Doppler

Accurate estimates of the location of the source of a Doppler shift in a patient are difficult to achieve with continuous-wave Doppler. The source is obviously somewhere along the extended axis of the ultrasound transducer (direction of sonation), but exactly where is uncertain. To localize the source of the Doppler shift, a method known as range gating may be used. In this method the sending transducer is shut off, and a preset interval of time, the first range gate, is allowed to elapse before the return signal is processed to determine the frequency shift. The Doppler shift is then analyzed until a second range gate is encountered. With the assumption of an average speed of sound, the range gate settings define a specific Doppler sampling region over a particular range of depth. The time between the two gates determines the extent of the sampling region, and variation of the elapsed time before the first gate determines the sampling depth. With a speed of sound for soft tissue of 1540 m/sec, a range factor of 13 μs of elapsed time corresponds to 1 cm of depth. This range factor takes into account the “round trip” of 2 cm that sound travels (i.e., from the transducer to the reflector and back again) for each centimeter of depth.

Example 22-4

Range gates are set at 39 and 45 μsec with respect to a transmitted pulse of ultrasound. Determine the dimensions of the Doppler sampling region within soft tissue.

$$\frac{39 \mu\text{sec (1st gate)}}{13 \mu\text{sec/cm}} = 3 \text{ cm from the transducer}$$

$$\frac{45 \mu\text{sec (2nd gate)}}{13 \mu\text{sec/cm}} = 3.5 \text{ cm from the transducer}$$

The Doppler sampling region begins at a depth of 3 cm and extends to a depth of 3.5 cm. We could state that there is a 0.5-cm sampling region centered at a depth of 3.25 cm. The width of the sampling region is roughly equal to the ultrasound beam width at that depth.

In Doppler systems that are used in conjunction with imaging, the position of the sampling volume is usually indicated by cursors positioned on the display. The ultrasound computer performs the electronic equivalent of the calculations in Example 22-4 to determine where the cursors should appear on the image.

Range gating requires that the ultrasound be started and stopped (i.e., pulsed) during the signal acquisition process so that the time to echo may be determined unambiguously. The use of pulsed Doppler not only allows range gating but also permits imaging pulses to be interspersed with Doppler pulses. Thus a real-time B-mode image of the patient may be obtained from every other pulse. In continuous-wave Doppler there is, by definition, no time in which to send imaging pulses, because there is no gap in the Doppler process.

A simplified diagram of a pulsed Doppler ultrasound system is shown in Margin Figure 22-5. The main differences between the pulsed Doppler system in this illustration and the continuous-wave system of Margin Figure 22-4 are that the pulsed system includes range gates and a transmit–receive switch similar to that in B-mode systems. Units that intersperse Doppler and imaging pulses incorporate components found in standard B-mode systems with both Doppler and B-mode components connected to the voltage generator or oscillator so that the sending of Doppler and imaging pulses can be coordinated.

Spectral Analysis

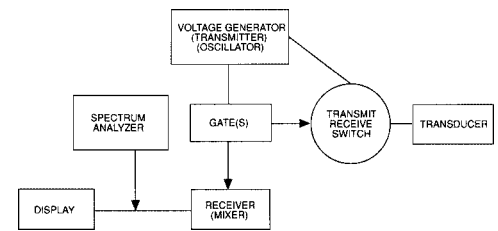
The Doppler-shift signal contains a wealth of information. One of the fundamental challenges of Doppler ultrasound is to display that information in a useful fashion. Doppler shift in a blood vessel is produced by the flow of red blood cells (RBCs). However, within a single vessel there is an assortment (distribution) of RBC velocities. Also, the velocity of any given cell varies with time according to the cardiac cycle. Three methods for visual display of information are discussed here: frequency spectrum, spectral trace, and color mapping.

Frequency Spectrum

A frequency spectrum (Margin Figure 22-6) is a histogram in which the horizontal axis depicts Doppler-shift frequency and the vertical axis shows the relative contribution of each frequency to the observed Doppler shift. For simplicity, the frequency spectrum can be thought of as a histogram of the relative distribution of RBC velocities in a blood vessel. The peak of the spectrum shows the Doppler shift associated with the greatest number of RBCs. The frequency spectrum is a “snapshot” depicting the conditions of flow in the sampling volume at a particular moment in time. To determine temporal variations, a display such as the flow plot is required.

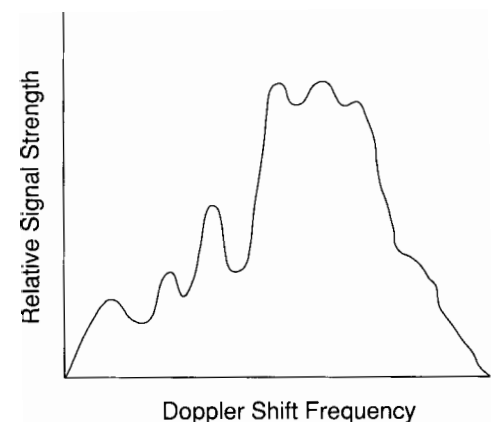
Spectral Trace

A spectral trace (Margin Figure 22-7) is a graph of frequency versus time that depicts the manner in which the value of Doppler shift varies with time. The value indicated may be the average, minimum, or maximum Doppler shift, and a distribution of values may be indicated by a vertical bar. The spectral trace may be displayed as a Doppler-shift frequency trace or as a velocity trace. The velocity trace is obtained by using the center frequency of the probe and the sonation angle measured by the operator to convert the Doppler-shift frequency scale a velocity scale (see Example 22-2). Audible sound may be broadcast by speakers as the trace is displayed in real time

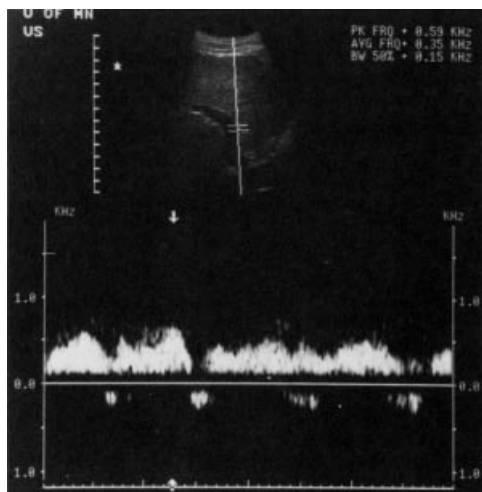


MARGIN FIGURE 22-5
Diagram of a pulsed Doppler unit.

There are techniques for determining velocity in ultrasound images that do not use the Doppler frequency shift effect. In “time domain” analysis, the return signals along scan lines are compared in a process known as “autocorrelation analysis” to determine if a signal anywhere in the image in one frame appears at a different location in another frame. This information is then color-coded using any of the techniques that are used in color Doppler.



MARGIN FIGURE 22-6
Doppler-shift frequency spectrum showing the relative contribution of frequencies to the Doppler shift during a brief interval of time.



MARGIN FIGURE 22-7

Doppler spectral trace showing Doppler shift as a function of time. The distribution of Doppler shifts at any given time is the vertical “thickness” of the trace. Thus, the Doppler spectral trace shows changes in the frequency spectrum over time.

In color Doppler, the most commonly used color scale codes flow away from the transducer as blue and flow toward the transducer as red. When such a “blue away red toward” (BART) display is used, it is sometimes necessary to remind observers that the colors are not necessarily correlated with the oxygenation (venous or arterial) status of the blood.

by using a “moving pen” or “strip chart” type of display. Cardiac information such as electrocardiographic (ECG) tracings may also be displayed to allow the viewer to coordinate features in the spectral trace with timing of the cardiac cycle. Some systems also allow the user to display a frequency spectrum for a selected moment of time during display of the spectral trace.

Color Mapping

Both frequency spectra and spectral traces may be displayed along with B-mode images by dividing the display screen into separate “windows” (compartments). Cursors on the image indicate the Doppler sampling volume. It is possible to indicate flow directly within the image by color-coding the pixels. In color mapping or color-flow Doppler, pixels in the B-mode image that correspond to regions in the patient where flow is present are assigned hues (colors such as red, blue, green) according to an arbitrary color scale. For example, red may indicate a large positive Doppler shift, orange a smaller positive shift, yellow a small negative shift, and so on.

Alternatively, the color scale may involve only two hues. One of the most common two-hue systems uses red for a positive Doppler shift and blue for a negative Doppler shift. The magnitude of the Doppler shift is encoded by the “saturation” of the color. Saturation is the degree to which the hue is present. A totally unsaturated color is white. It emits a broad distribution of all frequencies of visible light. Higher saturation implies that more of the frequencies in the spectrum are of a single frequency, interpreted by our retina as a purer color. The saturation encoding in color Doppler is usually arranged such that a greater shift (higher velocity either toward or away from the transducer) is encoded as a less saturated hue. For example, a small positive Doppler shift would be indicated by a dark red, whereas a larger positive Doppler shift would be indicated by a lighter, less saturated red.

Whatever color scale is used in color Doppler, it is desirable to avoid color-coding the very slow velocities of tissues that occur in the patient as pliable tissues are pushed by the transducer and as the transducer itself is moved. Detection of such motions is not usually the purpose of the exam. Thus, most color scales do not assign color to very small Doppler shifts and display a black band at the center of the color scale indicator. The width of the black band may be changed by the operator to vary the range of velocities that will be color-coded.

The large variety of color scales that could be used in color Doppler, as well as the variation in perception of color from one individual to another, is an intriguing problem for optimization of visual perception.

Quantitative Use of Doppler

Various attempts have been made to correlate information obtained from Doppler displays, particularly spectral traces, with pathology. Simple measures such as the peak systolic or end-diastolic average velocities are probably too crude to differentiate normal and disease states, in part because of the wide range of normal values in a population and also because of the difficulties inherent in obtaining quantitative values for velocity. It is possible to construct ratios of features of spectral traces so that conversion factors such as the angle of sonation are the same for both features and therefore cancel out. With the symbol A used for peak systolic and B for end-diastolic frequency shift, some indices that have been used with varying degrees of success include A/B ,⁵ $(A - B)/\text{mean}$,⁶ and $A - B/A$ (known as Pourcelot’s or the resistivity index).⁷ Still, attempts to provide definitive diagnoses on the basis of indices alone have not been generally successful. Such methods appear more useful for longitudinal studies than for specific diagnoses.⁸

LIMITATIONS OF DOPPLER SYSTEMS

Pulse Repetition Frequency

The use of pulsed Doppler limits the maximum velocity that can be detected. Each pulse reflected from the RBCs is a sample of the frequency shift. The greater the rate of sampling or pulsing, the better the measurement of the shift. Information theory suggests that an unknown periodic signal must be sampled at least twice per cycle to determine even rudimentary information such as the fundamental frequency.⁴ Thus the rate of pulsing [the pulse repetition frequency (PRF)] of pulsed Doppler must be at least twice the Doppler-shift frequency produced by flow. If not, the frequency shift is said to be “undersampled”—that is, sampled so infrequently that the frequency reported by the instrument is erroneously low, an artifact known as “aliasing.”

Example 22-5

A stenosis produces a high-speed jet in a vessel. The maximum velocity of RBCs in the stenosis is 80 cm/sec. Find the minimum PRF that must be used to avoid aliasing for pulsed Doppler at 7.5 MHz. Assume that the angle of sonation is 0 degrees.

The Doppler shift is

$$\begin{aligned}\Delta\nu &= \frac{2v_0 v \cos\theta}{c} \\ &= \frac{2(7.5 \times 10^6 \text{ sec}^{-1})(80 \text{ cm/sec})(1)}{1.54 \times 10^5 \text{ cm/sec}} \\ &= 7.8 \text{ kHz}\end{aligned}\quad (22-1)$$

The minimum PRF required to prevent aliasing is twice the Doppler-shift frequency, or 15.6 kHz.

There is a practical limit to the PRF of an ultrasound system. To obtain information at a given depth in the patient, the system must “wait and listen” over the time required for sound to travel from the transducer to that depth and back again. If an echo returns after another pulse has been sent, it is difficult to determine which travel time to use in determining the range. Because the travel time required for sound waves in soft tissue is 13 $\mu\text{sec}/\text{cm}$ of depth, the maximum PRF for a given total depth of imaging [field of view (FOV)] can be computed.

In color Doppler, aliasing is characterized by colors from one end of the color spectrum appearing adjacent to colors from the opposite end. For example, an abrupt transition from blue to red with no dark band between the two colors signals aliasing. A dark band would indicate momentary cessation of flow which would have to occur if there was a change in flow direction.

Example 22-6

Find the field of view (FOV) if the PRF of 15.6 kHz calculated from Example 22-5 is used for imaging.

If the PRF is 15.6 kHz, then the wait-and-listen time between pulses, the pulse repetition period (PRP), is its reciprocal.

$$\begin{aligned}\text{PRP} &= \frac{1}{\text{PRF}} = \frac{1}{15.6 \times 10^3 \text{ sec}^{-1}} = 6.41 \times 10^{-5} \text{ sec} \\ &= 64.1 \mu\text{sec}\end{aligned}$$

The field of view (FOV) is then

$$\text{FOV} = \frac{64.1 \mu\text{sec}}{13 \mu\text{sec}/\text{cm}} = 4.9 \text{ cm}$$

Clearly, the PRF required to measure the maximum flow in a high-speed jet is not routinely available in imaging systems because it would severely limit the field of view.

In echocardiography, the appearance of very rapidly moving blood that spurts from heart valves that do not close properly (known as “jets”) may be enhanced in color flow Doppler. One technique that enhances jets is to color code signals with an extremely high Doppler shift in a color that is markedly different from the traditional red and blue (e.g., green or yellow). Then, when a jet appears in a real time image, a spurt of bright color is immediately apparent.

To avoid aliasing, most pulsed Doppler systems have a “high-PRF” option that permits operation at PRFs that may exceed those practical for imaging. An alternative is to switch to continuous-wave Doppler because it is not limited in the same way that pulsed Doppler is limited by the sampling process. In some cases, it may be possible to switch to a lower-frequency transducer so that the Doppler shift is lower. Several techniques to circumvent the PRF requirement have been proposed but are still in the research stages. These include pulse-coding schemes and the use of nonperiodic pulses.^{9,10}

Spectral Broadening

There are many barriers to interpreting Doppler shift as a measure of a fundamental parameter such as velocity. Just as other imaging modalities are limited by noise, Doppler ultrasound is also limited by extraneous information or noise. This extraneous information may cause a “spectral broadening” of the range of Doppler-shift frequencies displayed by the unit. Spectral broadening compromises the interpretation of the width of the frequency spectrum as an indication of the range of RBC velocities. Spectral broadening can also be caused by other factors that may be lumped together as deterministic errors (see Chapter 11). These include the bandwidth or range of frequencies in the transmitted signal, the presence of adjacent vessels or flow within the fringes of the sampling volume, and tissue and wall motion (this low-velocity, low-frequency component is usually eliminated by electronic filtering known as “wall filtering”).

Nondeterministic or random spectral broadening may be caused by electronic noise resulting from random fluctuations in electronic signals.

PROBLEMS

- *22-1. Estimate the frequency shift for a 10-MHz ultrasound source moving toward a stationary detector in water ($c = 1540$ m/sec) at a speed of 5 cm/sec. Is the frequency shifted to a higher or lower value?
- *22-2. Estimate the frequency shift for an object moving at a speed of 10 cm/sec away from a 10-MHz source and detector occupying the same location.
- 22-3. What is meant by a beat frequency?
- 22-4. Identify some of the major medical applications of Doppler systems.
- *22-5. Find the minimum pulse repetition frequency required to prevent aliasing when measuring a velocity of 5 cm/sec with pulsed Doppler ultrasound at 5 MHz for a sonation angle of zero degrees. How would this PRF change if:
 - a. The angle of sonation was increased?
 - b. The center frequency was increased?

*For those problems marked with an asterisk, answers are provided on p. 493.

SUMMARY

- The Doppler equation is $\Delta v = 2v_0 \frac{v}{c} \cos \theta$.
- As the Doppler angle increases from zero to 90 degrees, the Doppler shift decreases and errors in estimation of the angle cause greater inaccuracies in velocity estimation.
- Volumetric flow Q , average velocity v , and area A are related as follows:

$$Q = vA$$

- The main components of a Doppler ultrasound imager include
 - Transducer
 - Transmit/receive switch
 - Voltage generator (transmitter or oscillator)
 - Gate system

- Receiver (mixer)
- Spectrum analyzer
- Display
- A Doppler spectral trace is a plot that shows the variation in time of a brightness modulated Doppler spectra.
- In color Doppler, the magnitude of frequency shift is indicated by saturation and the specific colors indicate direction of flow.
- Spectral broadening may indicate that a range of velocities exists in the sample volume. It may also be caused by
 - Bandwidth of the transmitted pulse
 - Signal from adjacent vessels
 - Tissue and vessel wall motion
 - Electronic noise

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FUNDAMENTALS OF MAGNETIC RESONANCE

OBJECTIVES 356

INTERACTION OF NUCLEI WITH A STATIC MAGNETIC FIELD 356

ROTATION AND PRECESSION 356

INTERACTION OF NUCLEI WITH A RADIO FREQUENCY WAVE:
NUTATION 357

INDUCTION OF A MAGNETIC RESONANCE SIGNAL
IN A COIL 358

QUANTUM MECHANICAL INTERPRETATION 359

BULK MAGNETIZATION 360

RELAXATION PROCESSES: T1 AND T2 361

RELAXATION TIMES (T1 AND T2) FOR BIOLOGIC MATERIALS 363

PROBLEMS 364

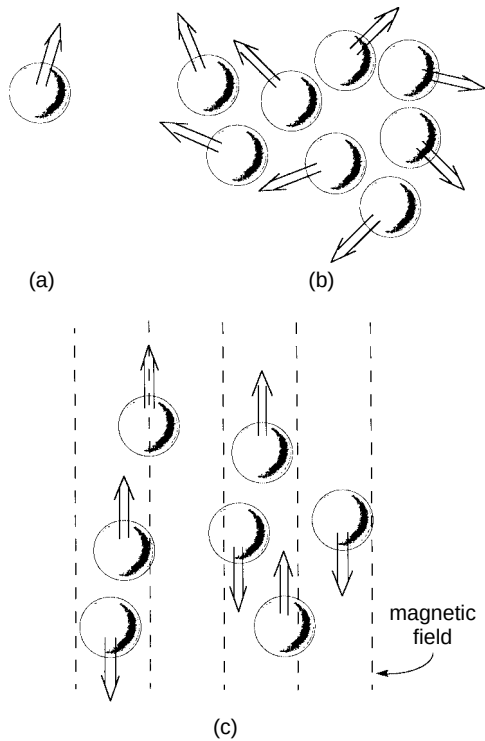
SUMMARY 364

REFERENCES 364

OBJECTIVES

After completing this chapter, the reader should be able to:

- Explain the phenomenon of precession of nuclei in a static magnetic field.
- Describe the phenomenon of nutation and explain how it is responsible for causing variation in the strength of the MR signal.
- Trace the steps involved in reception of the MR signal beginning with insertion of the patient into the magnet.
- Solve the Larmor equation to determine resonance frequency.
- Give the quantum mechanical explanation for magnetic resonance.
- Describe the behavior of bulk magnetization of a sample in the presence of static and radio frequency magnetic fields.
- Define T1 and T2 relaxation.
- Give an equation for exponential relaxation of spins after application of a radio-frequency pulse has ended.
- Explain the behavior of relaxation times as the strength of the static magnetic field is increased.



MARGIN FIGURE 23-1

A: Proton magnetic moment direction is indicated by arrow. **B:** In a typical material, magnetic moments are oriented randomly. **C:** If a magnetic field is applied, magnetic moments align themselves along the direction of the field. Note that some are parallel, while others are antiparallel.

In 1933, Stern and associates discovered that the proton had a magnetic moment and would therefore interact with a magnetic field.^{1,2} Six years later, one of Stern's students, Rabi, designed an apparatus to examine the interaction of magnetic nuclei with time-varying magnetic fields.³ He found that a certain frequency of the magnetic field was required to obtain the strongest interaction with an atom or molecule and that this frequency varied for each type of atom. Rabi demonstrated the principle of magnetic resonance (MR) and linked resonance frequencies to specific nuclei. He went on to investigate the magnetic forces exerted upon protons and neutrons. Rabi received the Nobel prize for this work in 1944.

Certain nuclei have magnetic properties as discussed in Chapter 2. By the use of magnetic fields and radio-frequency (RF) waves, these properties can be exploited to yield information about biological materials containing such nuclei.

INTERACTION OF NUCLEI WITH A STATIC MAGNETIC FIELD

This discussion of MR first presents the “classical interpretation” of the behavior of nuclear magnetic moments by using the hydrogen nucleus (i.e., a single proton) as a model. In the classical interpretation the position of the hydrogen nucleus can be specified with any desired degree of precision, and its movements are assumed to be continuous and completely predictable. Each proton behaves as a small magnet with a magnetic moment that has both magnitude and direction (Margin Figure 23-1). In a typical sample of hydrogen-containing material (such as the human body), the magnetic moments of the individual hydrogen nuclei are oriented in random directions. If a strong magnetic field is applied to the sample, the nuclei “align” their magnetic moments with the direction of the magnetic field in a manner similar to a compass needle aligned with the earth's magnetic field. The earth's magnetic field (0.5 gauss) is not strong enough to bring protons in a tissue sample into alignment. The field supplied by an MR system (e.g., 20,000 gauss) is strong enough to produce alignment.

ROTATION AND PRECESSION

In addition to aligning with a magnetic field, a magnetic moment also precesses about the field. Precession is easily demonstrated in rotating objects (another reason why we infer the property of “spin” for protons). A spinning top, for example, will “wobble” about a vertical axis defined by the earth's gravitational field (Margin Figure 23-2). This wobbling motion is precession.

Precession is a type of motion that is distinct from rotation. Rotation is the spinning of an object about its axis (an imaginary line through the center of mass of the object). The rapid spin of a top that causes its surface to blur is rotation. Precession is a “second-order” motion. It is the “rotation of a rotating object.”

If you push against a ball, it will roll in the direction in which it was pushed. If you push against a gyroscope (a gyroscope is simply a top surrounded by a stationary

framework), the gyroscope will move in a perpendicular direction (90 degrees) to the direction of the push. If you push it toward the north, it will move to the east or west depending on the direction of its spin (clockwise or counterclockwise) (Margin Figure 23-3).

In the macroscopic world, rotating objects have the property of angular momentum that causes them to behave as gyroscopes and tops. Nuclei react to forces in the microscopic world just as objects with angular momentum respond to forces in the macroscopic world. Protons and other subatomic particles are assumed to rotate about their axes and are described as having “spin.”

Precession results from the interaction of forces with a rotating object. These forces and the resultant motions are diagramed in Margin Figure 23-4 for a gyroscope, where the force is gravity, and for a proton, where the force is a magnetic field.

The frequency f of precession of a proton in units of megahertz (10^6 cycles or rotations per second) depends upon its gyromagnetic ratio γ (in megahertz per tesla) and the strength B (in tesla, T) of the static magnetic field. This relationship is described by the Larmor equation.

$$f = \gamma B \quad (23-1)$$

Gyromagnetic ratios for a number of nuclei are given in Table 23-1.

Example 23-1

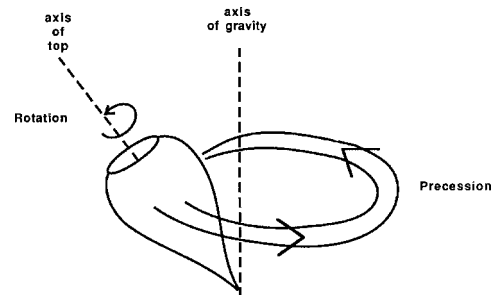
Find the resonance frequency for protons in a 2-tesla magnetic field.

$$\begin{aligned} f &= \gamma B \\ &= \left(42.6 \frac{\text{MHz}}{\text{T}} \right) (2.0 \text{ T}) \\ &= 85.2 \text{ MHz} \end{aligned} \quad (23-1)$$

INTERACTION OF NUCLEI WITH A RADIO FREQUENCY WAVE: NUTATION

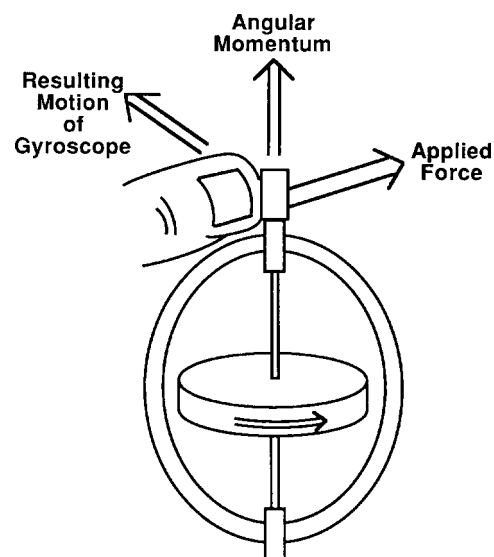
There is a “third-order” property of an object that is rotating (first-order property) and precessing (second-order property). This property, called nutation, is the result of forces that rotate *with* the precession of the object.

When force is applied to an object with angular momentum, the object tends to move at right angles to the force. If we try to speed up the precession of a gyroscope by using a finger to push it in a circle (Margin Figure 23-5), we will not affect the



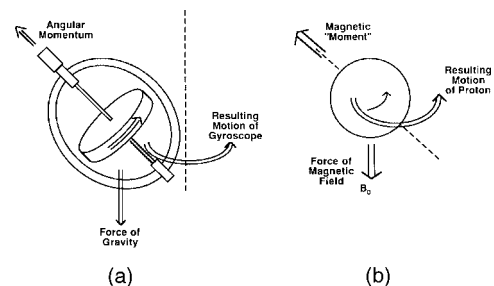
MARGIN FIGURE 23-2

Motions of a spinning top. Rotation or spin of the top about its own axis is first-order motion. Precession of the top about the vertical axis (axis of gravity) is second-order motion.



MARGIN FIGURE 23-3

Spinning object demonstrates a property known as angular momentum, shown as an arrow along the axis of the spinning object. When force is applied to an object having angular momentum, the resulting motion is at right angles to the force.



MARGIN FIGURE 23-4

Diagram of forces acting to cause precession. Angular momentum and gravity interact to cause precession of a gyroscope. Magnetic moment and a magnetic field interact to cause precession of a proton.

The frequency of precession is known as the *Larmor frequency*.

TABLE 23-1 Magnetic Resonance Properties of Selected Nuclei

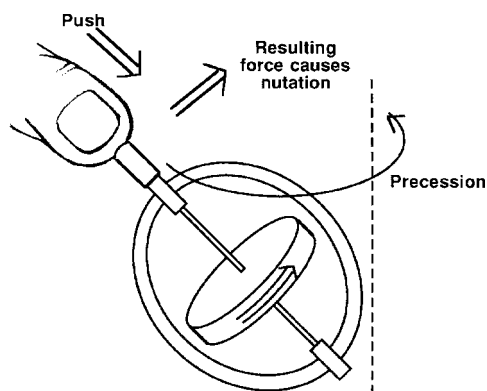
Nucleus	Natural Abundance (%)	Gyromagnetic Ratio ^a (MHz/T)	Sensitivity ^b
¹ H	99.98	42.58	100.00
¹³ C	1.11	10.71	1.59
¹⁹ F	100.00	40.05	83.30
²³ Na	100.00	11.26	9.25
³¹ P	100.00	17.23	6.63
³⁹ K	93.10	1.99	0.05

^aThe gyromagnetic ratio listed here is the same value listed in some texts as $\gamma/2\pi$, where γ is the gyromagnetic ratio in units of radians per tesla. The convention in this text is to assume that γ is given in units of megahertz per tesla, with the factor of 2π already included.

^bRelative sensitivity (signal strength as compared with hydrogen) for the same number of nuclei at constant field strength as a percentage of sensitivity for ¹H.

Source: Data from James, T. L. *Nuclear Magnetic Resonance in Biochemistry*. New York, Academic Press, 1975.

The emission of radio signals from nuclei was demonstrated independently by Bloch⁴ and Purcell⁵ in 1945. By the late 1940s, several groups had established that the electrical charges of electrons involved in chemical bonds caused small, precise shifts in the frequency of these signals. The frequency shifts were found to be characteristic for individual molecules. This finding identified nuclear magnetic resonance (NMR) as a method for identifying minute amounts of chemicals in unknown samples and led to the development of NMR in the 1950s as a standard tool for research in analytical chemistry.⁶



MARGIN FIGURE 23-5

Attemption to push a gyroscope in the direction of precession causes the gyroscope to change its angle of precession. Change in the angle of rotation is referred to as nutation.

Rotation: Cyclical motion of an object about an axis.

Precession: Compound motion of a rotating object about an axis other than its axis of rotation. The angle between the axis of rotation and the axis of precession is known as the *angle of precession*.

Nutation: Change in the angle of precession.

precessional speed. Instead, the *angle* of precession will change. That is, if we push in the direction of precession, the gyroscope precesses at a greater angle until it finally lies flat on the table. This result is best explained by a diagram that is drawn as if we were precessing along with the object. Such a “rotating-frame-of-reference” diagram is shown in Margin Figure 23-6. This diagram depicts a precessing object responding to a force by moving at a right angle to the force, thereby changing its *angle* of precession. This change in angle, called nutation, is a third-order circular motion (after rotation and precession).

In magnetic resonance of protons, the force that causes nutation is provided by a second magnetic field that varies with time. Colloquially, we could describe this field as follows: Each time the proton’s magnetic moment “comes around” during precession, the second magnetic field is “switched on” (Margin Figure 23-7). As the magnetic moment rotates away, the magnetic field is “switched off.” More precisely, the “ons” and “offs” are the peaks and valleys of a continuously varying magnetic field.

Example 23-1 shows that the Larmor equation [Eq. (23-1)] may be used to predict the frequency of precession of a proton in a magnetic field. The range of static field strengths employed in clinical MRI, 0.1 to 3.0 tesla, corresponds to precessional frequencies of 4.3 to 129 MHz for hydrogen nuclei. The time-varying magnetic field of an electromagnetic wave with a frequency in the megahertz range is a suitable source for inducing nutation. When this wave has a frequency that matches the precession of protons in a particular magnetic field, it is said to be in “resonance.” This is the origin of the term *magnetic resonance*. Appropriate frequencies are in the FM radio portion of the electromagnetic spectrum. Thus, we refer to the use of RF pulses in MR.

■ INDUCTION OF A MAGNETIC RESONANCE SIGNAL IN A COIL

A changing magnetic field can induce a current in a loop of conducting wire. This principle is known as Faraday’s law, or the law of electromagnetic induction. A proton has a magnetic moment and therefore acts as a small magnet. Protons that precess so that their magnetic fields intersect the plane of a nearby coil will induce an electrical current in the coil. This current is the MR “signal” induced in the receiver coil (Margin Figure 23-8).

The MR signal is greatest for precession of magnetic moments in a plane perpendicular to the plane of the receiver coil. The imaginary plane perpendicular to the receiver coil is known as the “transverse” or “xy” plane. If the magnetic moments are not precessing entirely within the transverse plane, a signal of reduced magnitude (i.e., a “weaker” signal) is induced in the coil. If the magnetic moments are perpendicular to the transverse plane (i.e., parallel to the static magnetic field), then no signal will be induced in the coil. In summary, it is the “component” of the magnetization in the transverse plane that is responsible for the MR signal.

The number of magnetic moments precessing near the coil also influences the strength of the MR signal. With other factors constant, the strength increases linearly with the number of magnetic moments. The magnetic moments that are used for MRI at the present time are those of protons. Signals from soft tissue in volumes on the order of a cubic millimeter can be resolved in MRI.

Example 23-2

Calculate the number of hydrogen nuclei in a cubic millimeter of water.

From the mass of a cubic millimeter of water, the number of water molecules and the number of protons can be computed.

The density of water is 1 g/cm^3 .

A cubic millimeter of water has a mass of

$$(1\text{ mm}^3) \left(10^{-3} \frac{\text{cm}^3}{\text{mm}^3} \right) \left(1 \frac{\text{g}}{\text{cm}^3} \right) = 0.001\text{ g}$$

A gram molecular weight, or mole, of water is the mass in grams of pure water that contains Avogadro's number, 6.02×10^{23} , of molecules. The molecular weight of water is 18 g/mol. The number of water molecules in 0.001 g is

$$\frac{(0.001 \text{ g})}{(18 \text{ g/mol})} \left(6.02 \times 10^{23} \frac{\text{molecules}}{\text{mol}} \right) = 3.34 \times 10^{19} \text{ molecules.}$$

However, each molecule contains two hydrogen nuclei. Therefore, the total number of hydrogen nuclei in a cubic millimeter of water is

$$\left(2 \frac{\text{H nuclei}}{\text{molecule}} \right) (3.34 \times 10^{19} \text{ molecule}) = 6.68 \times 10^{19} \text{ H nuclei}$$

Note that we do not count the eight protons in each oxygen atom in this calculation because those nuclei do not have the same Larmor frequency as hydrogen.

The maximum MR signal that may be induced in a receiver coil depends linearly upon the number of protons available. Hence doubling either the volume or the proton density (protons per cubic millimeter) of the sample doubles the MR signal.

■ QUANTUM MECHANICAL INTERPRETATION

The preceding discussion follows the “classical interpretation” of the behavior of magnetic moments in the presence of magnetic fields and radio waves. This interpretation is useful for teaching, but it is not entirely correct. In the region of the atom, where distances are small and the “discreteness” of energy levels and angular momentum is apparent, it is not possible to predict the exact behavior of individual atoms with certainty.⁷ Nuclei possess only certain discrete values of quantities such as angular momentum and energy; they cannot possess intermediate values of these variables. Properties such as angular momentum and energy are said to be quantized; that is, they are available only in specific values that are “allowed” by empirical rules. Therefore, the picture of smooth nutation of individual nuclei as if they were tiny tops gradually falling onto a tabletop is not strictly true.

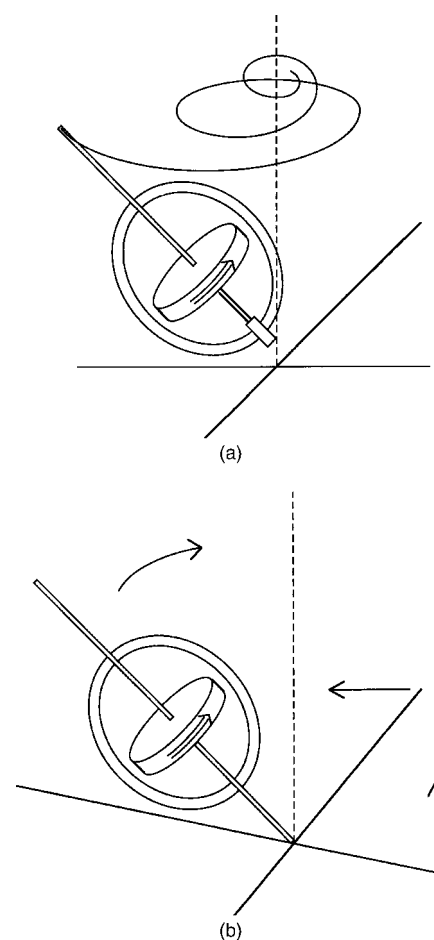
Another important distinction must be made between the quantum mechanical and the classic interpretations of MR. In quantum mechanics, instead of radio “waves,” we refer to photons or packets of electromagnetic energy. For example, in the quantum mechanical interpretation of a system of spins (such as protons) in a magnetic field, there are two possible “energy states,” either parallel or antiparallel to the magnetic field. The protons having magnetic moments aligned with the magnetic field have slightly less energy (are in a lower energy state) than do protons with magnetic moments opposing the magnetic field (Margin Figure 23-9). A photon with an energy equal to the energy difference between the two states can promote, or “flip,” protons from the lower to the higher energy state. In the classic model, the frequency of the radio wave that interacts in resonance is given by the Larmor equation [Eq. (23-1)]. In the quantum mechanical model, the energy of the photon is given by the equation

$$E = hf \quad (23-2)$$

where E is the energy of the photon; h is Planck's constant, 4.14×10^{-15} eV-sec; and f is the Larmor frequency in Hertz, or cycles per second. This equation permits a calculation of the photon energy required to cause transitions between low and high energy states.

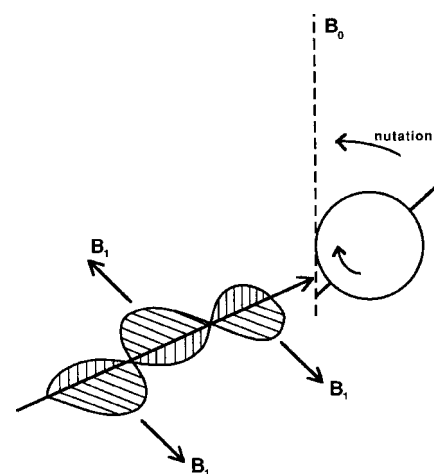
Example 23-3

Find the energy difference between low and high energy states of protons in a 1-tesla magnetic field.



MARGIN FIGURE 23-6

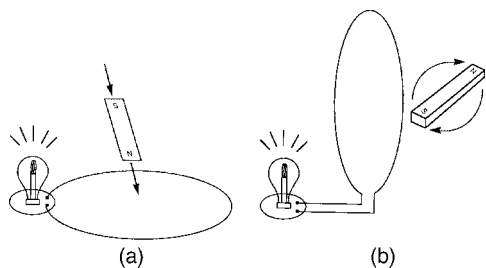
Nutation of a gyroscope shown in a stationary frame of reference (A) and a rotating frame of reference (B). Rotating frame of reference shows motion of the gyroscope as if the observer were precessing along with the gyroscope.



MARGIN FIGURE 23-7

Nutation of a precessing proton caused by application of an electromagnetic wave having intermittent magnetic field components (B_1). Motion is shown in the rotation frame of reference.

There are a number of magnetic resonance phenomena. Magnetic resonance effects involving the spin and orbital magnetic moments of an atom's electrons are actually thousands of times more powerful than the nuclear magnetic resonance effect used in magnetic resonance imaging. To date, no application in imaging the human body has been found for electron resonance phenomena. The effect described in this chapter is more accurately known as nuclear paramagnetic resonance. If an accurate physical description was the only consideration, the imaging described in this chapter should be abbreviated NPMRI.



MARGIN FIGURE 23-8

A: Magnet that is in motion near a loop of conducting wire induces an electrical current in the wire, causing the light bulb to glow. **B:** Magnet that rotates near a loop of conducting wire induces an alternating current in the wire, causing the light bulb to glow (or flash on and off if the rate of rotation of the magnet is low).

Werner Heisenberg, in the early 1970s, reflected on discussions among physicists in the 1920s on the subject of using terms such as “energy,” “magnetization,” and “reality” when describing quantum systems:

“It was found that if we wanted to adapt the language to the quantum theoretical mathematical scheme, we would have to change even our Aristotelian logic. That is so disagreeable that nobody wants to do it: . . . We have learned that language is a very dangerous instrument to use, and this fact will certainly have its repercussions in other fields, but this is a very long process which will last through many decades I should say.”

(From Buckley, P., Peat, F. D., Peat, D. F. (eds.), *Glimpsing Reality: Ideas in Physics and the Link to Biology*, revised edition (March 1996), University of Toronto Press.

First, the precessional frequency of the protons is computed from the Larmor equation:

$$\begin{aligned} f &= \gamma B \\ &= \left(42.6 \frac{\text{MHz}}{\text{T}} \right) (1\text{T}) \\ &= 42.6 \text{ MHz} = 42.6 \times 10^6 \text{ sec}^{-1} \end{aligned} \quad (23-1)$$

Next the energy difference is computed from the Larmor frequency:

$$\begin{aligned} E &= hf \\ &= 4.14 \times 10^{-15} \text{ eV} \cdot \text{sec} (42.6 \times 10^6 \text{ sec}^{-1}) \\ &= 1.76 \times 10^{-7} \text{ eV} \end{aligned} \quad (23-2)$$

Thus a photon energy of $1.76 \times 10^{-7} \text{ eV}$ is required. As expected, this photon is part of the RF portion of the electromagnetic spectrum.

BULK MAGNETIZATION

The classic and quantum mechanical interpretations of MR phenomena can be reconciled when a sample of macroscopic size consisting of millions of atoms is considered. In this case, the classical laws govern the composite, or “bulk,” magnetization of the sample. It is a cornerstone principle of physics that as the quantum mechanical equations governing the behavior of individual atoms are averaged over “large” numbers of atoms, the classical equations of motion emerge. Thus, the statements about precession, nutation, and induction of signals in a coil are true of bulk magnetization. When a radio wave is applied to a sample at the resonance frequency, the bulk magnetization precesses at the Larmor frequency (Margin Figure 23-10). The angle of precession (nutation angle) is determined by the intensity of the radio wave and the amount of time over which it is applied.

When considering a collection of spins, as in a macroscopic sample, an important issue to include is whether the individual magnetic moments tend to “work together” (i.e., point in the same direction) or cancel each other (i.e., point in random directions). Before a radio wave is applied, the nuclei are precessing at nearly the same frequency, but their transverse components tend to cancel each other because they are rotating “out of phase.” Hence the bulk magnetization is zero in the transverse plane. When the radio wave (RF pulse) is applied, the spins are brought into synchrony, and the individual nuclear magnetic moments reinforce each other to produce a strong bulk magnetic moment. The ability of an RF pulse to bring protons into phase is a consequence of applying a force that acts in resonance with a physical system. As an example, consider using a long board to simultaneously push several people who are on swings.* Assume that you will push at the resonance frequency. If the people are already swinging on their own in random phases as you approach them with the board to apply “pulses” to the system, you will bump some at just the right part of the swing cycle to give them an effective push. Others will be missed or will bump against the board at the wrong part of the cycle. If you persist over a number of cycles, those who are out of phase will be forced into phase, and finally the force will be applied to all components of the system in resonance. This technique is analogous to what happens when an RF pulse is applied to a system of precessing nuclear spins. As the nuclei begin to rotate in phase, the transverse component of the bulk magnetization becomes apparent.

*In this physical system, resonance frequency is determined by the length of the ropes supporting the swings and the mass of the people.

TABLE 23-2 Relaxation Times (Mean $\pm$ SD) in Milliseconds for Various Tissues at 1 Tesla Static Magnetic Field Strength (42.6 MHz)

Tissue	T1			T2		
Skeletal muscle	732	$\pm$	132	47	$\pm$	13
Liver	423	$\pm$	93	43	$\pm$	14
Kidney ^a	589	$\pm$	159	58	$\pm$	24
Adipose	241	$\pm$	68	84	$\pm$	36
Brain						
Gray matter	813	$\pm$	138	101	$\pm$	13
White matter	683	$\pm$	116	92	$\pm$	22

^aAverage of medulla and cortex.Source: Data from Bottomley, P. A., Foster, T. H., Argersinger, R. E., et al. *Med. Phys.* 1984; **11**: 524–547.

RELAXATION PROCESSES: T1 AND T2

We have seen that when an RF pulse is applied to a sample, the bulk magnetization may be nutated into the xy plane and induce an MR signal in a receiver coil positioned perpendicular to the xy plane. When the radio wave is switched off, the signal decays away. This decay is the result of the return of protons to the state that existed before the radio wave was applied. This return is termed *relaxation* of the protons. There are two basic relaxation processes at work in the sample. Both processes account for the observed decay of the MR signal.

One relaxation process involves a return of the protons to their original alignment with the static magnetic field. This process, called *longitudinal* or *spin-lattice relaxation*, is characterized by a time constant T1 (Margin Figure 23-11). The term spin-lattice refers to the interaction of the protons (spins) with their surroundings (the “lattice” or network of other spins). This interaction causes a net release of energy to the surroundings as the protons return to the lower energy state of alignment.

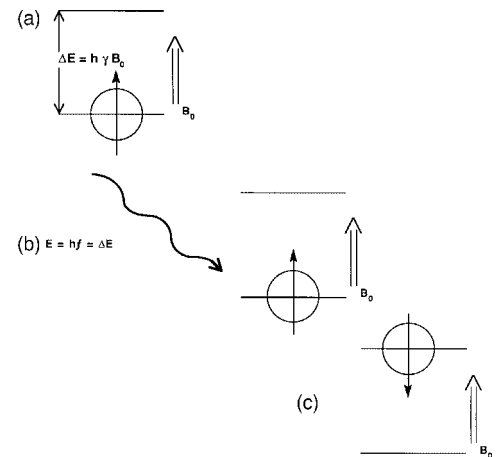
The other relaxation process is a loss of synchrony of precession among the protons. Before a radio wave is applied, the precessional orientation of the protons is random. The application of a radio wave brings the protons into synchronous precession, or “in phase.” When the radio wave is switched off, the protons begin to interact with their neighbors and give up energy in random collisions. In so doing, they revert to a state of random phase. As the protons revert to random orientation, the bulk signal decreases because the magnetic moments tend to cancel each other. This process is called *transverse* or *spin-spin relaxation* and is characterized by a time constant T2 (Margin Figure 23-11).

In a patient undergoing MRI, both longitudinal and transverse relaxation processes occur at the same time. The transverse (T2) relaxation time is always shorter than the longitudinal (T1) relaxation time. That is, magnetic moments dephase faster than they move into alignment with the static magnetic field (Margin Figure 23-11). For typical biological materials, T1 may be on the order of several hundred milliseconds while T2 is a few tens of milliseconds (Table 23-2).¹⁶

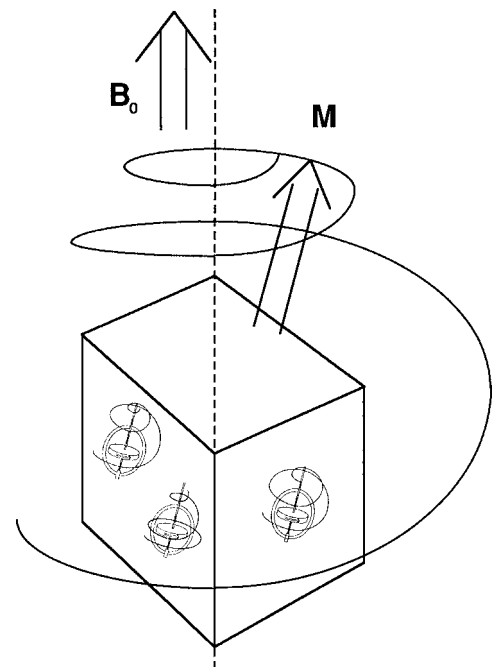
Relaxation (decay) of the MR signal is characterized by exponential expressions analogous to those used to describe radioactive decay, absorption of photons, and growth of cells in culture (see Appendix I). For longitudinal relaxation, the expression

$$S = S_0 e^{-t/T1} \quad (23-3)$$

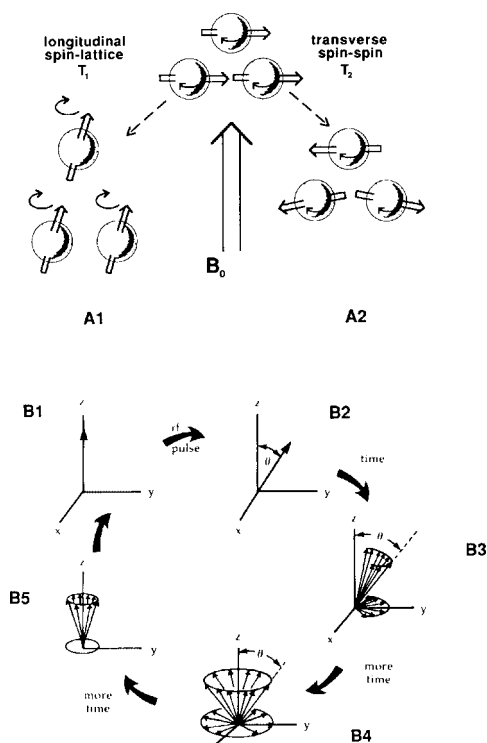
reveals that the MR signal S decreases exponentially in time from the signal S_0 that was present immediately following an RF pulse. The value of S_0 is influenced by factors such as the number of protons in the sample, length of time that the radio wave was applied to the sample, sensitivity of the receiver coil, and overall sensitivity of the electronics.

**MARGIN FIGURE 23-9**

In the quantum mechanical model of magnetic resonance, protons parallel to (aligned with) the magnetic field B_0 exist in a lower energy state than protons that are antiparallel. A photon having an energy equal to the difference in energy between the two states can change a proton in the lower energy state to the higher energy state.

**MARGIN FIGURE 23-10**

Bulk magnetization of a sample of spins represents the net result of the precession and nutation of the individual spins within a sample.

**MARGIN FIGURE 23-11**

A: Two relaxation processes in a sample of nuclear spins. There are several terms for each process.

A1: Longitudinal relaxation occurs as the spins return to alignment with the static magnetic field, B_0 . **A2:** Transverse relaxation occurs as the spins precess "out of phase." **B:** spin-spin relaxation diagramed in the rotating frame of reference. **B1:** Magnetic moment of the sample is aligned with the magnetic field. **B2:** Immediately after an RF pulse, the magnetic moment of the sample can be represented by a single vector. **B3:** As the magnetization vector begins to break up or dephase as a result of localized nonuniformities in the applied field, components of the vector begin to fan out in the xy plane. **B4:** When there are an equal number of components in all directions in the xy plane, the components cancel one another and the MR signal disappears. **B5:** As time passes, the cone representing the precessing but dephased magnetic moment continues to narrow because of spin-lattice relaxation. **B1:** Finally the magnetic moment once again is realigned with the applied field. (From Hendee, W. R., and Morgan, C. R. *West. J. Med.* 1984; **141**:491–500. Used with permission.)

Example 23-4

If a material has a longitudinal relaxation time constant $T_1 = 200$ msec, what will the MR signal S be 200 ms after an RF pulse has flipped (nutated) the bulk magnetization into the transverse plane?

$$S = S_0 e^{-t/T_1} \quad (23-3)$$

for $t = T_1 = 200$ msec,

$$\begin{aligned} S &= S_0 e^{-200/200} \\ &= S_0 e^{-1} \\ &= S_0(0.37) \end{aligned}$$

This example illustrates that the longitudinal time constant T_1 may be defined as the time required for the signal to decrease to 37% of its original value by the process of longitudinal (spin-lattice) relaxation.

For transverse relaxation, governed by the time constant T_2 , we obtain

$$S = S_0 e^{-t/T_2} \quad (23-4)$$

where T_2 is defined as the time required for the MR signal to decrease to 37% of its original value by the process of transverse (spin-spin) relaxation. Because T_1 and T_2 are independent processes both acting to reduce the signal S_0 , Eq. (23-3) and (23-4) may be combined into

$$S = S_0 e^{-t/T_1} e^{-t/T_2} \quad (23-5)$$

Example 23-5

A sample of liver tissue has a longitudinal relaxation time constant T_1 of 500 msec and a transverse relaxation time constant T_2 of 40 ms. What will the MR signal be 40 msec after an RF pulse? 500 msec after?

$$S = S_0 e^{-t/T_1} e^{-t/T_2} \quad (23-5)$$

At $t = 40$ msec,

$$\begin{aligned} S &= S_0 e^{-40/500} e^{-40/40} \\ &= S_0(0.92)(0.37) \\ &= S_0(0.34) \end{aligned}$$

At 40 msec the signal is 34% of its value immediately following an RF pulse.

At $t = 500$ msec,

$$\begin{aligned} S &= S_0 e^{-500/500} e^{-500/40} \\ &= S_0(0.37)(3.7 \times 10^{-6}) \\ &= S_0(1.38 \times 10^{-6}) \end{aligned}$$

At 500 msec the signal is little more than a millionth of its value immediately after the RF pulse.

Note that during this time, both longitudinal and transverse relaxation processes act to reduce the magnitude of the MR signal. However, the effect of T_2 dominates, particularly at longer time intervals. If T_1 is long compared with T_2 as in this example (and as is the case for many tissues of interest in MRI), the effect of T_1 is overshadowed by the effect of T_2 . However, if T_1 is about the same as T_2 , as for nonviscous liquids, then both relaxation processes are important influences on the MR signal.

In the next chapter, special RF pulsing techniques are introduced to vary the influence of T_1 and T_2 upon the MR signal that is finally recorded. With special pulsing techniques the MR signal may actually appear to increase in strength due